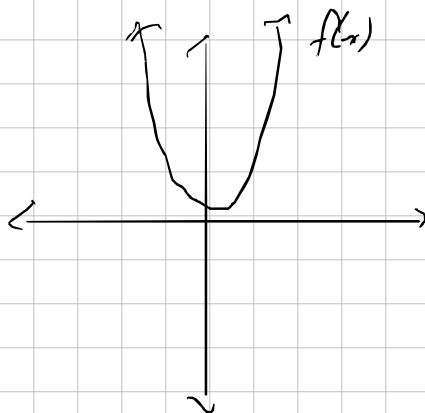


M/433 Introduction to Optimization

Textbook: Optimal Control.

Calc I Review

$$f(x) = x^4$$



$$\frac{df(x)}{dx} = f'(x) = 0$$

Fermat's Theorem

$$f'(x) = 0 = 4x^3 \rightarrow x^* = 0$$

↑
"stationary point"

$$\frac{d^2f}{dx^2} f''(x) = 12x^2 \rightarrow f''(x^* = 0) = 0 \leftarrow \text{inconclusive}$$

If second derivative test is inconclusive,

Taylor series expansion

$$f(x) \approx f(x^*) + \frac{f'(x^*)}{1!} (x - x^*) + \frac{f''(x^*)}{2!} (x - x^*)^2 + \dots$$

necessary condition

$$f'''(x) = 24x \rightarrow f'''(x^* = 0) = 0$$

$$f^{(4)}(x) = 24 \rightarrow \underline{f^{(4)}(x^* = 0) = 24}$$

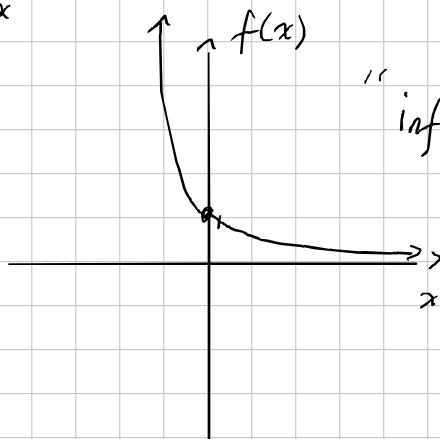
$$f(x) \approx f(x^*) + \frac{f'''(x^*)}{4!} (x - x^*)^4 = x^4$$

look for $x = x^* \pm \delta$

$$\left. \begin{array}{l} x^* + \delta \rightarrow f(x) > f(x^*) \\ x^* - \delta \rightarrow f(x) > f(x^*) \end{array} \right\} f(x) > f(x^*) \text{ in both cases must be a minimum.}$$

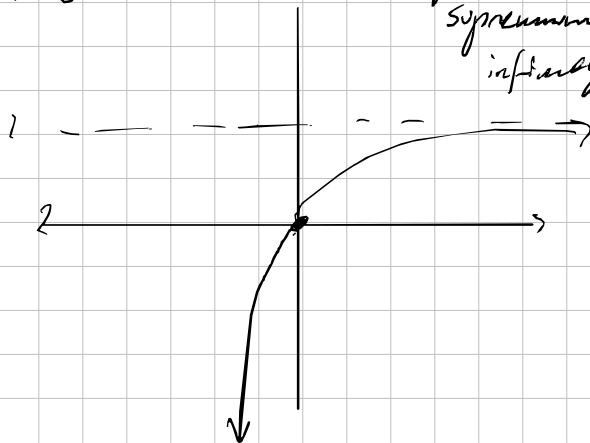
Necessary condition: $f'(x^*) = 0$
 Sufficient condition: $f''(x^*) > 0$ positive semi-definite
 for a minimum.

$$f(x) = e^{-x}$$



"infimum" = 0
 infinitely approaches 0

$$f(x) = 1 - e^{-x}$$



"supremum" = 1
 infinitely approaches 1

$\min f(x), x \in I$ for interval $I [a, b]$

$\min f(x), x \in D$ for domain $D (-\infty, \infty)$

$\min L(\bar{u})$

↑
cost

↑ control variables: $m \times 1$

Fenchel's Theorem

$$\rightarrow \frac{\partial L}{\partial \bar{u}} = \bar{0} \rightarrow (1 \times m)$$

necessary
conditions for
optimality

rows of numerator
col of denominator

$$\frac{\partial^2 L}{\partial \bar{u}^2} = \begin{bmatrix} \frac{\partial^2 L}{\partial u_1^2} & \frac{\partial^2 L}{\partial u_1 \partial u_2} & \dots & \frac{\partial^2 L}{\partial u_1 \partial u_m} \\ \frac{\partial^2 L}{\partial u_2 \partial u_1} & \frac{\partial^2 L}{\partial u_2^2} & & \vdots \\ \vdots & & \ddots & \vdots \\ \frac{\partial^2 L}{\partial u_m \partial u_1} & & & \frac{\partial^2 L}{\partial u_m^2} \end{bmatrix}$$

$(m \times m)$

Fenchel's Theorem: $\frac{\partial^2 L}{\partial u_m \partial u_1} = \frac{\partial^2 L}{\partial u_1 \partial u_m} \rightarrow \text{Symmetric}$

Symmetric & Symmetric matrix $\rightarrow$ Hessian Matrix
for $\frac{\partial^2 L}{\partial \bar{u}} > 0 \rightarrow$ positive semi-definite

all eigenvalues have to be ≥ 0 for

$$\frac{\partial^2 L}{\partial \bar{u}^2} \geq 0.$$

iff all eigenvalues are > 0 , "positive definite"
and you have a minimum.

$$\text{Ex: } L(\bar{u}) = u_1^2 - 2u_1u_2 + 4u_2^2$$

$$\frac{\partial L}{\partial \bar{u}} = \bar{0} = \left[\frac{\partial L}{\partial u_1}, \frac{\partial L}{\partial u_2} \right] = [2u_1 - 2u_2, -2u_1 + 8u_2]$$

2 eqns 2 unk.

$$\begin{bmatrix} 2 & -2 \\ 2 & 8 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\text{stationary pt. } \begin{bmatrix} u_1^* \\ u_2^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\frac{\partial^2 L}{\partial \bar{u}^2} = \begin{bmatrix} 2 & -2 \\ -2 & 8 \end{bmatrix} \quad \text{iff Hessian not constant, then} \rightarrow u = \bar{u}^*$$

Now need eigenvalues of the matrix

$$\det(A - \lambda \text{ Eigenvalue}) = 0$$

$$[A]\vec{v} = \lambda \vec{v}$$

$\nearrow$ eigenvectors $\nearrow$ eigenvalues

$$\det \begin{bmatrix} 2-\lambda & -2 \\ -2 & 8-\lambda \end{bmatrix} = (2-\lambda)(8-\lambda) - 4 = 0$$

$$\lambda^2 - 10\lambda + 12 = 0$$

$$\lambda = \frac{10 \pm \sqrt{52}}{2} = 5 \pm \sqrt{13}$$

both positive

$$\frac{\partial^2 L}{\partial \vec{u}^2} \rightarrow \text{positive definite}$$

Ex: $L(\vec{u}) = u_1^2 - 4u_1u_2 + 3u_2^2$

$$\frac{\partial L}{\partial \vec{u}} = [2u_1 - 4u_2, -8u_1 + 12u_2]$$

$$\therefore \vec{u} = \vec{0}$$

$$\frac{\partial^2 L}{\partial \vec{u}^2} = \begin{bmatrix} 2 & -8u_2 \\ -8u_2 & -8u_1 + 12u_2^2 \end{bmatrix} \xrightarrow{\vec{u} = \vec{0}} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\det \begin{bmatrix} 2-\lambda & 0 \\ 0 & -\lambda \end{bmatrix} = 0 \rightarrow (-\lambda)(2-\lambda) = 0 \rightarrow \lambda = 2, 0$$

positive semi-definite.

Equality Constraints

Let $L(\bar{p})$ subject to $\bar{f}(\bar{p}) = \bar{0}$
 $\uparrow$
"parameters"
 $(n \times 1)$

"states" $\bar{x} (n \times 1)$ to satisfy the $n \times 1$ constraints.

"controls" $\bar{u} (m \times 1)$

parameters $\bar{p} = \begin{bmatrix} \bar{x} \\ \bar{u} \end{bmatrix}$
size $[(n+m) \times 1]$

$$dL = 0 = \frac{\partial L}{\partial \bar{x}} \delta \bar{x} + \frac{\partial L}{\partial \bar{u}} \delta \bar{u}$$

$(1 \times n) (n \times 1) \quad (1 \times m) (m \times 1)$

scalar result

$$d\bar{f} = \bar{0} = \left(\frac{\partial \bar{f}}{\partial \bar{x}} \right) \delta \bar{x} + \frac{\partial \bar{f}}{\partial \bar{u}} \delta \bar{u}$$

$(n \times n) (n \times 1) \quad (n \times m) (m \times 1)$

$(n \times 1)$

Jacobian Matrix.

$$\frac{\partial \bar{f}}{\partial \bar{x}} \delta \bar{x} = - \frac{\partial \bar{f}}{\partial \bar{u}} \delta \bar{u}$$

$$\delta \bar{x} = - \left(\frac{\partial \bar{f}}{\partial \bar{x}} \right)^{-1} \frac{\partial \bar{f}}{\partial \bar{u}} \delta \bar{u}$$

dL is a linear combination of the sum of the total change of $f, \bar{f}$.

$$\underbrace{\frac{\partial L}{\partial \bar{x}}}_{(1 \times n)} + \underbrace{\bar{\lambda}^T \frac{\partial \bar{f}}{\partial \bar{x}}}_{(1 \times n)(n \times n) \rightarrow (1 \times n)} = \bar{0} \quad ; \quad \bar{\lambda} = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{bmatrix}$$

Lagrange
multipliers
or adjoint
variables.

(1 × n)

$$\underbrace{\frac{\partial L}{\partial \bar{u}}}_{(1 \times m)} + \underbrace{\bar{\lambda}^T \frac{\partial \bar{f}}{\partial \bar{u}}}_{(1 \times n)(n \times m) \rightarrow (1 \times m)} = \bar{0}$$

(1 × m)

$$\bar{f}(\bar{x}, \bar{u}) = \bar{0} \quad (n \times 1)$$

Then we have $2n+m$ equations and $2n+m$ unknowns.

$$\begin{aligned} &\bar{x} (n \times 1) \\ &\bar{\lambda} (n \times 1) \\ &\bar{u} (m \times 1) \end{aligned}$$

$$H = x^2 + y^2 + z^2 + \lambda_1 (5x + 7y + 2z - 4)$$

$$\frac{\partial H}{\partial x} = 0 = 2x + 5\lambda_1,$$

$$\frac{\partial H}{\partial z} = 0 = [2y + 7\lambda_1, 2z + 2\lambda_1]$$

4 eqns
4 unk.

$$\text{Constraint: } 5x + 7y + 2z - 4 = 0$$

Solve:

$$\begin{bmatrix} 2 & 0 & 0 & 5 \\ 0 & 2 & 0 & 7 \\ 0 & 0 & 2 & 2 \\ 5 & 7 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ \lambda_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \\ \lambda_1 \end{bmatrix} \approx \begin{bmatrix} 0.2564 \\ 0.3590 \\ 0.1026 \\ -0.1026 \end{bmatrix}$$

$$\hookrightarrow = \frac{1}{39} \begin{bmatrix} 10 \\ 14 \\ 4 \\ -4 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 7 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 4$$

Ans Sol

$$\nabla L = \frac{\partial L}{\partial \vec{p}} = \left[\frac{\partial L}{\partial x} \quad \frac{\partial L}{\partial y} \quad \frac{\partial L}{\partial z} \right] = [2x, 2y, 2z]$$

Second order

$$L(\vec{p}) \approx L(\vec{p}^0) + \frac{\partial L}{\partial \vec{p}} \delta \vec{p} + \frac{1}{2} \frac{\partial^2 L}{\partial \vec{p}^2} \delta \vec{p}^2 + \dots$$

$$H = L + \lambda^T \bar{f}$$

constraints = 0

Second order change

$$\begin{bmatrix} \delta \bar{x} & \delta \bar{u} \end{bmatrix} \begin{bmatrix} \frac{\partial^2 H}{\partial \bar{x}^2} & \frac{\partial^2 H}{\partial \bar{x} \partial \bar{u}} \\ \frac{\partial^2 H}{\partial \bar{u} \partial \bar{x}} & \frac{\partial^2 H}{\partial \bar{u}^2} \end{bmatrix} \begin{bmatrix} \delta \bar{x} \\ \delta \bar{u} \end{bmatrix}$$

$$\delta \bar{x} = - \left(\frac{\partial \bar{f}}{\partial \bar{x}} \right)^{-1} \frac{\partial \bar{f}}{\partial \bar{u}} \delta \bar{u}$$

$$\frac{\partial^2 L}{\partial \bar{u}^2} = \begin{bmatrix} - \left(\frac{\partial \bar{f}}{\partial \bar{x}} \right)^{-1} \frac{\partial \bar{f}}{\partial \bar{u}} & I_{m \times m} \end{bmatrix} \begin{bmatrix} \frac{\partial^2 H}{\partial \bar{x}^2} & \frac{\partial^2 H}{\partial \bar{x} \partial \bar{u}} \\ \frac{\partial^2 H}{\partial \bar{u} \partial \bar{x}} & \frac{\partial^2 H}{\partial \bar{u}^2} \end{bmatrix} \begin{bmatrix} - \left(\frac{\partial \bar{f}}{\partial \bar{x}} \right)^{-1} \frac{\partial \bar{f}}{\partial \bar{u}} \\ I_{m \times m} \end{bmatrix}$$

Shortcut:

$$\frac{\partial \bar{f}}{\partial \bar{x}} = \bar{f}_x, \quad \frac{\partial \bar{f}}{\partial \bar{u}} = \bar{f}_u, \quad \frac{\partial^2 H}{\partial \bar{x}^2} = H_{xx}, \quad \frac{\partial^2 H}{\partial \bar{x} \partial \bar{u}} = H_{xu}$$

$$\frac{\partial^2 L}{\partial \bar{u}^2} = \begin{bmatrix} \bar{f}_x^T & I_{m \times m} \end{bmatrix} \begin{bmatrix} H_{xx} & H_{xu} \\ H_{ux} & H_{uu} \end{bmatrix} \begin{bmatrix} \bar{f}_x \\ I_{m \times m} \end{bmatrix}$$

$$\bar{z} = - (\bar{f}_x)^{-1} \bar{f}_u$$

Ex: constraint is the plane: $5x + 7y + 2z - 4 = 0$

$$\bar{x} = x, \quad \bar{u} = \begin{bmatrix} y \\ z \end{bmatrix}$$

$$\frac{\partial f}{\partial x} = 5 \quad \frac{\partial f}{\partial u} = \bar{f}_u = \begin{bmatrix} 7 & 2 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{\text{size of num} \times \text{size of denom}} \quad 1 \times 2$

$$z = -(\bar{f}_x)^{-1} \bar{f}_u = -\left(\frac{1}{5}\right) \begin{bmatrix} 7 & 2 \end{bmatrix} = \begin{bmatrix} -7/5 & -2/5 \end{bmatrix}$$

$$H = x^2 + y^2 + z^2 + \lambda(5x - 7y + 2z - 4)$$

$$H_x = 2x + 5\lambda$$

$$H_u = [2y + 7\lambda, 2z + 2\lambda]$$

$$H_{xx} = 2$$

$$H_{xu} = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_{H_{xy} \quad H_{xz}}$

$$H_{ux} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$u \times x$

$$H_{uu} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$\frac{\partial^2 L}{\partial u^2} = \begin{bmatrix} -7/5 & 1 & 0 \\ -2/5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -7/2 & -5/2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$2 \times 3 \times 3 \times 2$

$$= \begin{bmatrix} -14/5 & 2 & 0 \\ -4/5 & 0 & 2 \end{bmatrix} \begin{bmatrix} -7/5 & -2/5 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{98}{25} + 2 & \frac{28}{25} \\ \frac{28}{25} & \frac{8}{25} + 2 \end{bmatrix} = \rightarrow$$

$$= \frac{1}{25} \begin{bmatrix} 148 & 28 \\ 28 & 58 \end{bmatrix} = \frac{2}{25} \begin{bmatrix} 74 & 14 \\ 14 & 29 \end{bmatrix}$$

Curvature of function on plane.

To determine sign of curvature:

Eigenvalues: $\frac{2}{25} [78, 25] > 0$. Both positive, positive definite.

$\mathcal{L}$ is minimum!

cubic equation should be p2.

Inequality constraints.

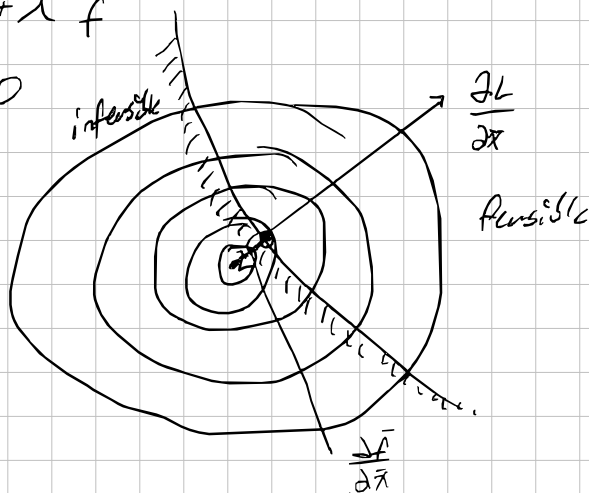
min $\mathcal{L}(p)$ subject to: $\bar{f} \leq \bar{0}$ (e.g. $x \geq 0$)
 $-x \leq 0$

1) inequality is "active": $f_i = 0, \lambda_i = \bar{\lambda}_i$, must be $\leq$

2) inequality is "inactive": $f_i < 0, \lambda_i = 0$

$$H = L + \bar{\lambda}^T \bar{f}$$

$$\bar{\lambda}^T \bar{f} = 0$$



$$\frac{\partial H}{\partial \bar{x}} = \bar{0} = \frac{\partial L}{\partial \bar{x}} + \lambda^T \frac{\partial \bar{f}}{\partial \bar{x}}$$

$\therefore \lambda_i \geq 0$ for all active constraints.

KKT conditions (Karush-Kuhn-Tucker)

1) inequality is "active" : $f_i = 0, \underline{\lambda_i} \geq 0$

2) inequality is "inactive" : $f_i < 0, \lambda_i = 0$

If the cost and the constraints are all linear,

$\hookrightarrow$ "linear programming problem"

$\hookrightarrow$ MATLAB "linprog"

$\hookrightarrow$ uses simplex method

Else anythly nonlinear:

$\hookrightarrow$ "nonlinear programming problem"

$\hookrightarrow$ MATLAB "fmincon"

Simplex Method.

$\rightarrow$ take the dot product between $\nabla L \left(\frac{\partial L}{\partial \bar{x}} \right)$ and each row of $\frac{\partial \bar{f}}{\partial \bar{x}}$

$\rightarrow$ select the n most negative constraints as active.

$\rightarrow$ solve the problem, confirm that all constraints, f_i are satisfied $\rightarrow$

→ and all i are ≥ 0

→ If true, solution found.

→ Else, release the most positive constraint from the dot products, move along the most negative constraint until another constraint is active.

→ Repeat until soln found.

"Fundamental Theorem of Linear Programming"

→ the solution lies @ the intersection of n of the constraints. No unboundedness.

Example: 800 empty bottles
225 bottles of 100% alcohol

wine is 20 proof (10% ABV)
whiskey is 80 proof (40% ABV)

Enough sugar for 600 bottles of wine
or 1200 bottles of whiskey

\$1 profit / bottle of wine

\$2 profit / bottle of whiskey

Maximize profit.

$$\text{Max profit} = \text{revenue} - \text{profit}$$

$$\text{min } L = - (x + 2y)$$

wine bottles: x
 whiskey bottles: y

Constraints:

$$f_1: x + y \leq 800 \rightarrow x + y - 800 \leq 0 \quad (\text{bottles})$$

$$f_2: \frac{x}{10} + \frac{2}{5}y \leq 225 \rightarrow x + 4y - 2250 \leq 0 \quad (\text{ABV})$$

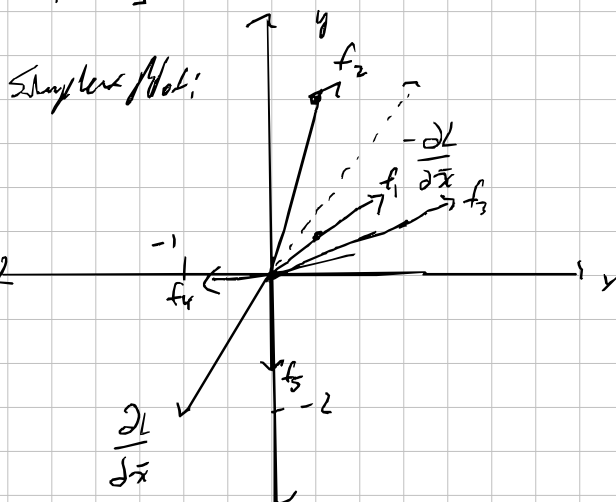
$$f_3: \frac{x}{600} + \frac{y}{1200} \leq 1 \rightarrow 2x + y - 1200 \leq 0 \quad (\text{sugar})$$

$$f_4: x \geq 0 \rightarrow -x \leq 0$$

$$f_5: y \geq 0 \rightarrow -y \leq 0$$

} (positive bottles)

$$\frac{\partial L}{\partial x} = [-1, -2]$$



$$\frac{\partial \bar{F}}{\partial \bar{x}} = \begin{bmatrix} 1 & 1 \\ 1 & 4 \\ 2 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$$

plot constraint gradient on
simplex plot.

size of $f \times$ size of x (5×2)

Most regular 2 constraints.

f_1 and f_2 are active.

f_3, f_4 , and f_5 are inactive.

$$\lambda_3^*, \lambda_4^*, \lambda_5^* = 0$$

$$\begin{bmatrix} \lambda_3^* \\ \lambda_4^* \\ \lambda_5^* \end{bmatrix} = \bar{0}$$

Now equality constraint prob.

$$H = L + \bar{\lambda}^T \bar{F}$$

$$H = -(x+2y) + \lambda_1 (x+y-800) + \lambda_2 (x+4y-2250)$$

$$\left\{ \begin{array}{l} \frac{\partial H}{\partial \bar{x}} = \bar{0} = [-1 + \lambda_1 + \lambda_2, -2 + \lambda_1 + 4\lambda_2] \\ x+y-800=0 \\ x+4y-2250=0 \end{array} \right.$$

Solving, $\lambda_1^* = \frac{2}{3}$, $\lambda_2^* = \frac{1}{3}$

$$\bar{\lambda} = \begin{bmatrix} 2/3 \\ 1/3 \\ 0 \\ 0 \\ 0 \end{bmatrix} \geq 0$$

$\therefore$ satisfies KKT conditions.

Solving 2 const.

$x^* = 316\frac{2}{3}$ when bottles

$y^* = 483\frac{1}{3}$ whisky bottles

Case 1

$x = 317$

$y = 483$

Case 2

$x = 316$

$y = 484$

$\rightarrow$ violates ABV
constraints
 $\downarrow$

does not satisfy t_2

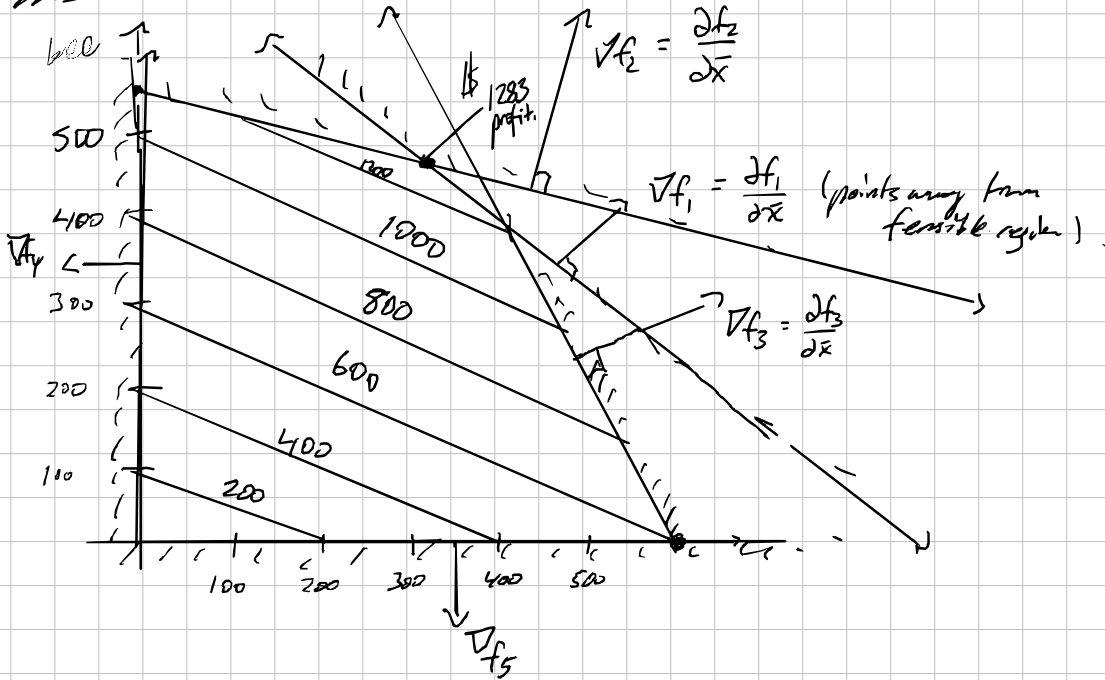
plug into constraints

Satisfies f_1, f_2

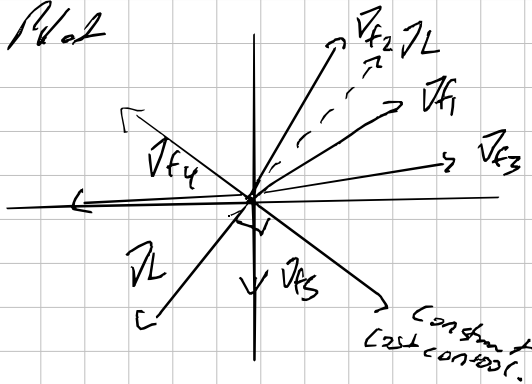
$$L = -(x + 2y)$$

$$\begin{aligned} \bar{f}: \quad f_1 &= x + y - 800 \leq 0 & \rightarrow y &\leq -x + 800 \\ f_2 &= x + 4y - 2250 \leq 0 & \rightarrow y &\leq -\frac{1}{4}x + 562.5 \\ f_3 &= 7x + y - 1200 \leq 0 & \rightarrow y &\leq -7x + 1200 \\ f_4 &= -x \leq 0 & \rightarrow & \\ f_5 &= -y \leq 0 & \rightarrow & \end{aligned}$$

Plot



Simpler Plot



$$L = -(x + 2y) = \underbrace{[-1 \ -2]}_f \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\text{linprog} : f^T x$$

$$\text{linprog}([[-1] \\ [-2]])$$

$\bar{f} :$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 4 \\ 2 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}}_A \begin{bmatrix} x \\ y \end{bmatrix} \leq \underbrace{\begin{bmatrix} 800 \\ 2250 \\ 1200 \\ 0 \\ 0 \end{bmatrix}}_b$$

$$-x_1 - 3x_2 + 3 \leq 0$$

$$[0 \ 3] \ [1 \ 0]$$

$$\text{Ex: } y \leq \frac{1}{2}x + 3 \quad f_1$$

$$y \leq -\frac{2}{3}(x-4) \quad f_2$$

$$x \geq -1 \quad f_3$$

$$y \geq 0 \quad f_4$$

min distance to $(2, 3)$

$$d = \sqrt{(x-2)^2 + (y-3)^2}$$

$$L = (x-2)^2 + (y-3)^2$$

$$H = (x-2)^2 + (y-3)^2 + \lambda_2 \left(\frac{2}{3}(x-4) + y \right)$$

1 constraint $\Rightarrow \bar{x} = [x]$
 1 control: $\bar{u} = [y]$

$$\frac{\partial H}{\partial x} = \frac{\partial H}{\partial x} = 2(x-2) + \frac{2}{3}\lambda_2 = 0$$

$$\frac{\partial H}{\partial y} = \frac{\partial H}{\partial y} = 2(y-3) + \lambda_2 = 0$$

$$\frac{\partial H}{\partial \lambda_2} = \frac{2}{3}(x-4) + y = 0$$

} 3 equations

$$\begin{bmatrix} 2 & 0 & \frac{2}{3} \\ 0 & 2 & 1 \\ \frac{2}{3} & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ \frac{8}{3} \end{bmatrix} \Rightarrow \begin{bmatrix} 16/13 \\ 24/13 \\ 30/13 \end{bmatrix} = \begin{bmatrix} x^* \\ y^* \\ \lambda_2^* \end{bmatrix}$$

$$\lambda^* = \begin{bmatrix} 0 \\ 30/13 \\ 0 \\ 0 \end{bmatrix} \neq 0 \rightarrow \text{substitute KKT conditions.}$$

$$2(p^*) = 1.923 \rightarrow \boxed{d^* = 1.39}$$

Second ORDER OPTIMALITY:

$$\frac{\partial^2 L}{\partial \bar{u}^2} = \begin{bmatrix} z^T & I_{m \times m} \end{bmatrix} \begin{bmatrix} H_{xx} & H_{xu} \\ H_{ux} & H_{uu} \end{bmatrix} \begin{bmatrix} z \\ I_{m \times m} \end{bmatrix}$$

$$z = -\left(\frac{\partial \bar{f}}{\partial \bar{x}}\right)^{-1} \frac{\partial \bar{f}}{\partial \bar{u}} = -\left(\frac{2}{3}\right)^{-1} \cdot 1 = -\frac{3}{2}$$

$$\frac{\partial \bar{f}}{\partial \bar{x}} = \frac{\partial f_2}{\partial x} = \frac{2}{3}$$

$$\frac{\partial \bar{f}}{\partial \bar{u}} = \frac{\partial f_2}{\partial y} = 1$$

$$\frac{\partial^2 L}{\partial \bar{u}^2} = \begin{bmatrix} -3/2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -3/2 \\ 1 \end{bmatrix} = 13/2 > 0 \quad \therefore \text{is minimum.}$$

Ex 3.17

$$f(x, y) = 9x^2 + 13y^2 + 18xy - 4$$

subject to $x^2 + y^2 + 2x \geq 16$

$$\rightarrow (x+1)^2 + y^2 \geq 17$$

circle centered at $(-1, 0)$ w/ rad $\sqrt{17}$.
outside of circle is feasible region.

$$\text{with } L = 9x^2 + 13y^2 + 18xy - 4$$

$\rightarrow$ graphically, constraint is active.

$$H = L + \lambda_1 \bar{f} = 9x^2 + 13y^2 + 18xy - 4 + \lambda_1 (17 - (x+1)^2 - y^2)$$

1 constraint $\rightarrow$ 1 slack

$$\rightarrow \bar{x} = x$$

$$\rightarrow \bar{u} = y$$

$$H_{\bar{x}} = 0 = \frac{\partial H}{\partial x} = 18x + 18y - 2\lambda_1(x+1)$$

$$H_{\bar{u}} = 0 = \frac{\partial H}{\partial y} = 26y + 18x - 2\lambda_1 y$$

$$H_{\bar{f}} = 0 = 17 - (x+1)^2 - y^2$$

} KKT eqns.

$$\frac{\partial^2 \mathcal{L}}{\partial u^2} = L_{uu} = \begin{bmatrix} z^T & I_m \end{bmatrix} \begin{bmatrix} H_{xx} & H_{xu} \\ H_{ux} & H_{uu} \end{bmatrix} \begin{bmatrix} z \\ I_m \end{bmatrix}$$

where $z = -(f_x)^{-1} f_u$

$$z = -(-2(x+1))^{-1} \cdot (-2y)$$

solve using x^* , y^*

$$z =$$

$$H_{xx} = 18 - 21, \xrightarrow{1=1}$$

$$H_{xu} = H_{ux} = 18$$

$$H_{uu} = 26 - 21, \xrightarrow{1=1}$$

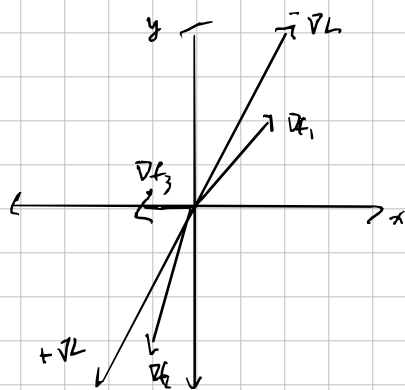
$$\max f(\bar{x}) = x_1 + 2x_2 \text{ subject to } \bar{f}$$

$$\bar{f}_1 = 3x_1 + 4x_2 \leq 12 \rightarrow 3x_1 + 4x_2 - 12 \leq 0$$

$$\bar{f}_2 = x_1 + 3x_2 \geq 3 \rightarrow -x_1 - 3x_2 + 3 \leq 0$$

$$\bar{f}_3 = x_1 \geq 0 \rightarrow -x_1 \leq 0$$

Linear problem, to determine active constraints, simplex method is used:



$$\mathcal{L}(\bar{p}) = -(x_1 + 2x_2) \quad \bar{x} = [x_1, x_2]$$

$$\frac{\partial \mathcal{L}}{\partial \bar{x}} = [-1 \ -2]$$

$$\frac{\partial \bar{f}}{\partial \bar{x}} = \begin{bmatrix} 3 & 4 \\ -1 & -3 \\ -1 & 0 \end{bmatrix}$$

$\nabla f_1, \nabla f_3$ must register $\rightarrow f_1, f_3$ is active constraints.

$$H = \mathcal{L} + \lambda \bar{f} = -x_1 - 2x_2 + \lambda_1 (3x_1 + 4x_2 - 12) + \lambda_2 (-x_1)$$

$$\frac{\partial H}{\partial \bar{x}} = \bar{0} = \begin{bmatrix} -1 + 3\lambda_1 - \lambda_2 \\ -2 + 4\lambda_1 \end{bmatrix}$$

$$\frac{\partial H}{\partial \lambda} = \bar{0} = \begin{bmatrix} 3x_1 + 4x_2 - 12 \\ -x_1 \end{bmatrix}$$

4 linear eqns
4 unknowns.

$$\begin{bmatrix} x_1^* \\ x_2^* \\ \lambda_1^* \\ \lambda_2^* \end{bmatrix} = \begin{bmatrix} 0 & 0 & 3 & -1 \\ 0 & 0 & 4 & -3 \\ 3 & 4 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 2 \\ 12 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 1/2 \\ 1/2 \end{bmatrix}$$

$$\lambda_2 = 0 \rightarrow \bar{\lambda} = \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix} \geq 0 \text{ is KKT conditions satisfied.}$$

$$\mathcal{L}^*(\bar{p}) = -(x_1 + 2x_2) = -6 \rightarrow \max f(\bar{x}) = 6.$$

Generally, 1 constraint active

$$f_2 = x_2 \leq 5 \rightarrow x_2 - 5 \leq 0$$

$$Z(\bar{x}) = (x_1 - 4)^2 + (x_2 - 8)^2$$

$$\bar{x} = x_1, \quad \bar{u} = x_2$$

$$H = L + \bar{\lambda} \bar{f} = (x_1 - 4)^2 + (x_2 - 8)^2 + \lambda_2 (x_2 - 5)$$

$$\frac{\partial H}{\partial \bar{x}} = \frac{\partial H}{\partial x_1} = 2(x_1 - 4) = 0$$

$$\frac{\partial H}{\partial \bar{u}} = \frac{\partial H}{\partial x_2} = 2(x_2 - 8) + \lambda_2 = 0$$

$$\frac{\partial H}{\partial \bar{\lambda}} = \frac{\partial H}{\partial \lambda_2} = x_2 - 5 = 0$$

} 3 eqns (linear)
3 unknown

$$\begin{bmatrix} x_1 \\ x_2 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} 8 \\ 16 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\lambda_1 = \lambda_3 = \lambda_4 = 0 ; \text{ all inactive}$$

$$\lambda_2 = 6$$

} $\bar{\lambda} \geq 0 \therefore$ 1st satisfied.

$$\mathcal{L}(\bar{p}) = 3x_1^2 - 2x_1x_2 + 5x_2^2 + 8x_2 \quad \bar{x} = x_1, \bar{u} = x_2$$

$$\bar{f} = f_1 = x_1^2 - x_2^2 + 8x_2 \geq 17$$

$$H = \mathcal{L} + \lambda \bar{f} = 3x_1^2 - 2x_1x_2 + 5x_2^2 + 8x_2 + \lambda_1 (17 - x_1^2 + x_2^2 - 8x_2)$$

$$\frac{\partial H}{\partial \bar{x}} = \frac{\partial H}{\partial x_1} = 6x_1 - 2x_2 - 2\lambda_1 x_1$$

$$\frac{\partial H}{\partial \bar{u}} = \frac{\partial H}{\partial x_2} = -2x_1 + 10x_2 + 2\lambda_1 x_2 + 8 - 8\lambda_1$$

$$\frac{\partial H}{\partial \lambda} = \frac{\partial H}{\partial \lambda} = 17 - x_1^2 + x_2^2 - 8x_2$$

} 3 nonlinear eqns
3 unknowns.

Solve in MATLAB

$$\begin{bmatrix} x_1^* \\ x_2^* \\ \lambda_1^* \end{bmatrix} = \begin{bmatrix} 2.9494 \\ 1.2247 \\ 2.5848 \end{bmatrix} \quad \lambda \geq 0 : \text{KKT satisfied.}$$

$$\frac{\partial^2 \mathcal{L}}{\partial u^2} = L_{uu} = \begin{bmatrix} z^T & I_m \end{bmatrix} \begin{bmatrix} H_{xx} & H_{xu} \\ H_{ux} & H_{uu} \end{bmatrix} \begin{bmatrix} z \\ I_m \end{bmatrix}$$

$$\text{where } z = -(f_x)^{-1} f_u$$

$$z = -(2x_1, \lambda_1)^{-1} \cdot (-2x_2, 8 - 8\lambda_1)$$

$$z = -0.940812$$

$$L_{uu} = \begin{bmatrix} -0.9408 & 1 \end{bmatrix} \begin{bmatrix} 6 - 2\lambda_1^* & -2 \\ -2 & 10 + 2\lambda_1^* \end{bmatrix} \begin{bmatrix} -0.940812 \\ 1 \end{bmatrix} = 19.6678$$

$\geq 0 \checkmark$
 $\therefore$ is a min

$$\text{but } \mathcal{L}_1^* = 2.5848$$

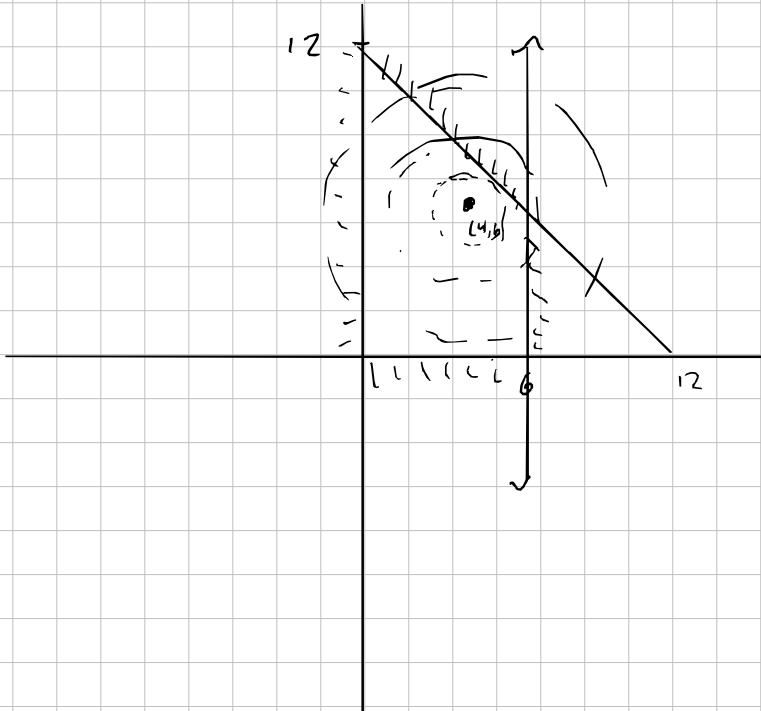
$$\min L = (x-4)^2 + (y-6)^2$$

$$f_1: x+y \leq 12 \quad \rightarrow \quad x+y-12 \leq 0$$

$$f_2: 6 \geq x \quad \rightarrow \quad x-6 \leq 0$$

$$f_3: x \geq 0 \quad \rightarrow \quad -x \leq 0$$

$$f_4: y \geq 0 \quad \rightarrow \quad -y \leq 0$$



2 active constraints: f_3, f_4
 f_1, f_2 inactive

$$\lambda_1 = \lambda_2 = 0$$

$$\bar{x} = [x, y] \quad \bar{u} = []$$

$$H = -(x-4)^2 + (y-6)^2 + \lambda_3 (x) + \lambda_4 (-y)$$

$$\frac{\partial H}{\partial \bar{x}} = \bar{0} = \begin{bmatrix} -2(x-4) - \lambda_3 \\ -2(y-6) - \lambda_4 \end{bmatrix}$$

$$\frac{\partial H}{\partial \bar{\lambda}} = \bar{0} = \begin{bmatrix} -x \\ -y \end{bmatrix}$$

$$\begin{bmatrix} x^* \\ y^* \end{bmatrix} = \bar{0} \rightarrow L^* = 52$$

$$\lambda_3^* = 8$$

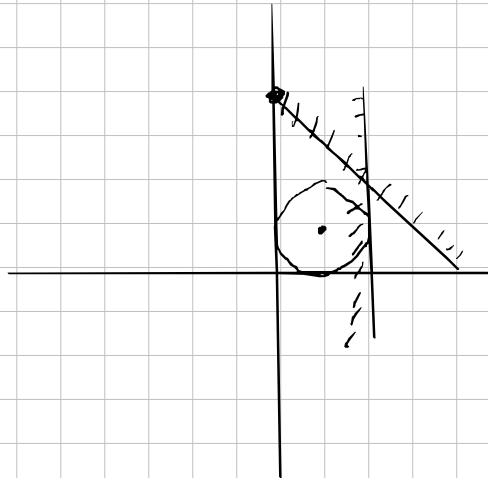
$$\lambda_4^* = 12$$

$$\bar{\lambda}^* = \begin{bmatrix} 0 \\ 0 \\ 8 \\ 12 \end{bmatrix} \geq 0 \quad \therefore \text{Schwefler Kkt.}$$

$$\text{min } L = (x_1 - 1)^2 + (x_2 - 1)^2$$

$$f_1: x_1 + x_2 - 4 \leq 0$$

$$f_2: 2 - x_1 \leq 0$$



f_2 is active, $\lambda_1 = 0$

$\bar{x} = x_1$, $\bar{u} = x_2$ must select this way

$$H = (x_1 - 1)^2 + (x_2 - 1)^2 + \lambda_2 (2 - x_1)$$

$$\frac{\partial H}{\partial \bar{x}} = \frac{\partial H}{\partial x_1} = 2(x_1 - 1) - \lambda_2 = 0$$

$$\frac{\partial H}{\partial \bar{u}} = \frac{\partial H}{\partial x_2} = 2(x_2 - 1) = 0$$

$$\frac{\partial H}{\partial \bar{L}} = \frac{\partial H}{\partial \lambda_2} = 2 - x_1 = 0$$

$$\begin{bmatrix} x_1^* \\ x_2^* \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$\lambda_2^* = 2$$

$$\bar{\lambda} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \geq 0 \quad \therefore \text{satisfies KKT.}$$

$$L_{uu} = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} H_{xx} & H_{xu} \\ H_{ux} & H_{uu} \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \textcircled{2}$$

$$z = -(f_{xx})^{-1} f_u = -(-1)^{-1} \cdot 0 = 0 > 0, \quad \therefore \text{is minimum.}$$

$$\max z = x_1 + 2x_2 \rightarrow \min L = -x_1 - 2x_2$$

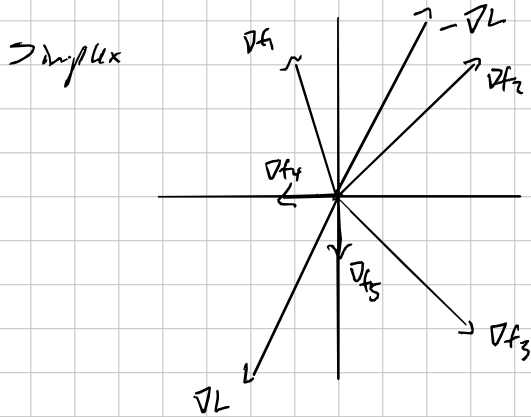
$$f_1: -x_1 + 3x_2 - 10 \leq 0$$

$$f_2: x_1 + x_2 - 6 \leq 0$$

$$f_3: x_1 - x_2 - 2 \leq 0$$

$$f_4: -x_1 \leq 0$$

$$f_5: -x_2 \leq 0$$



f_1, f_2 active from simplex plot.

$$\lambda_3 = \lambda_4 = \lambda_5 = 0 \quad (f_3 - f_5 \text{ inactive})$$

minimize "Bolza Cost Function"

$$J = \underbrace{\phi_0(\bar{x}_0, t_0)}_{\text{initial cost}} + \underbrace{\phi_f(\bar{x}_f, t_f)}_{\text{terminal cost}} + \underbrace{\int_{t_0}^{t_f} L(\bar{x}, \bar{u}, t) dt}_{\text{continuous cost}}$$

subject to: $\dot{\bar{x}} = \bar{f}(\bar{x}, \bar{u}, t)$

I.C. $\Psi_0(\bar{x}_0, t_0) = \bar{0}$

T.C. $\bar{\Psi}_f(\bar{x}_f, t_f) = \bar{0}$

$$J = \phi_0(\bar{x}_0, t_0) + \bar{v}_0^T \bar{\Psi}_0(\bar{x}_0, t_0) + \underbrace{\phi_f(\bar{x}_f, t_f) + \bar{v}_f^T \bar{\Psi}_f(\bar{x}_f, t_f) + \int_{t_0}^{t_f} (L(\bar{x}, \bar{u}, t) - \bar{\lambda}^T (\bar{f}(\bar{x}, \bar{u}, t) - \dot{\bar{x}})) dt}_{= \bar{0}}$$

$H(\bar{x}, \bar{u}, t)$

$\bar{\Phi}_f(\bar{x}_f, t_f)$

→ $\bar{\Phi}_0(\bar{x}_0, t_0)$

Altogether now:

$$J = \bar{\Phi}_0(\bar{x}_0, t_0) + \bar{\Phi}_f(\bar{x}_f, t_f) + \int_{t_0}^{t_f} [H(\bar{x}, \bar{u}, t) - \bar{\lambda}^T \dot{\bar{x}}] dt$$

$$\int_{t_0}^{t_f} -\bar{\lambda}^T \dot{\bar{x}} dt = -\bar{\lambda}^T \bar{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \dot{\bar{\lambda}}^T \bar{x} dt$$

$$\underbrace{u = -\bar{\lambda}^T \quad v = \bar{x}}_{du = -\dot{\bar{\lambda}}^T dt \quad dv = \dot{\bar{x}} dt}$$

$$\int u dv = uv - \int v du$$

$$\int_{t_0}^{t_f} -\dot{\lambda}^T \dot{\bar{x}} dt = -\lambda^T \bar{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \dot{\lambda}^T \bar{x} dt$$

$$J = \Phi_0(\bar{x}_0, t_0) + \Phi_f(\bar{x}_f, t_f) - \lambda^T(t_f) \bar{x}_f + \lambda^T(t_0) \bar{x}_0 + \int_{t_0}^{t_f} (H(\bar{x}, \bar{u}, t) + \dot{\lambda}^T \bar{x}) dt$$

$$\delta J = \frac{\partial J}{\partial \bar{x}} \delta \bar{x} + \frac{\partial J}{\partial \bar{u}} \delta \bar{u}$$

$$\delta J = \left[\frac{\partial \Phi_0}{\partial \bar{x}_0} + \lambda^T(t_0) \right] \delta \bar{x}_0 + \left[\frac{\partial \Phi_f}{\partial \bar{x}_f} - \lambda^T(t_f) \right] \delta \bar{x}_f + \int_{t_0}^{t_f} \left[\left(\frac{\partial H}{\partial \bar{x}} + \dot{\lambda}^T \right) \delta \bar{x} + \left(\frac{\partial H}{\partial \bar{u}} \right) \delta \bar{u} \right] dt = 0$$

~~To~~ guarantee a soln, all terms must equal zero. @ optimal soln.

$$\delta J = \underbrace{\left[\frac{\partial \Phi_0}{\partial \bar{x}_0} + \lambda^T(t_0) \right]}_{=0} \delta \bar{x}_0 + \underbrace{\left[\frac{\partial \Phi_f}{\partial \bar{x}_f} - \lambda^T(t_f) \right]}_{=0} \delta \bar{x}_f + \int_{t_0}^{t_f} \underbrace{\left[\left(\frac{\partial H}{\partial \bar{x}} + \dot{\lambda}^T \right) \delta \bar{x} \right]}_{=0} + \underbrace{\left[\left(\frac{\partial H}{\partial \bar{u}} \right) \delta \bar{u} \right]}_{=0} dt = 0$$

Euler-Lagrange Equations (Necessary Conditions for optimal calculus of variations problems)

$$\dot{\lambda}^T = -\frac{\partial H}{\partial \bar{x}} \quad \lambda^T(t_0) = -\frac{\partial \Phi_0}{\partial \bar{x}_0}$$

$$\frac{\partial H}{\partial \bar{u}} = 0 \quad \lambda^T(t_f) = \frac{\partial \Phi_f}{\partial \bar{x}_f}$$

typically $\bar{x}_0$ known $\rightarrow \lambda(t_0)$ are unknown

$\bar{x}_f$ unknown $\rightarrow \lambda(t_f)$ are known

Two point boundary value problem

$$\left\{ \begin{array}{l} \dot{\bar{x}} = f(\bar{x}, \bar{u}, t) \\ \dot{\lambda}^T = -\frac{\partial H}{\partial \bar{x}} \end{array} \right. \quad \left. \begin{array}{l} \bar{x}_0 \text{ known} \\ \lambda(t_f) \text{ known} \end{array} \right\} \leftarrow \begin{array}{l} \text{TPBVP} \\ \text{typical} \end{array}$$

"Hrd problems to solve"

in MATLAB: "bvp4c"

$$\frac{\partial}{\partial t} H(\bar{x}, \bar{u}, t) = \frac{\partial H}{\partial t} = \frac{\partial H}{\partial \bar{x}} \dot{\bar{x}} + \frac{\partial H}{\partial \bar{u}} \dot{\bar{u}} + \frac{\partial H}{\partial t}$$

$L + \bar{\lambda}^T \bar{f}$

@ optimal

$$\downarrow$$
$$(-\dot{\bar{\lambda}}^T) \dot{\bar{x}} + (0) \dot{\bar{u}} + \left(\frac{\partial L}{\partial t} + \dot{\bar{\lambda}}^T \bar{f} + \bar{\lambda}^T \frac{\partial \bar{f}}{\partial t} \right)$$

$\downarrow$
 $= \dot{\bar{f}}$

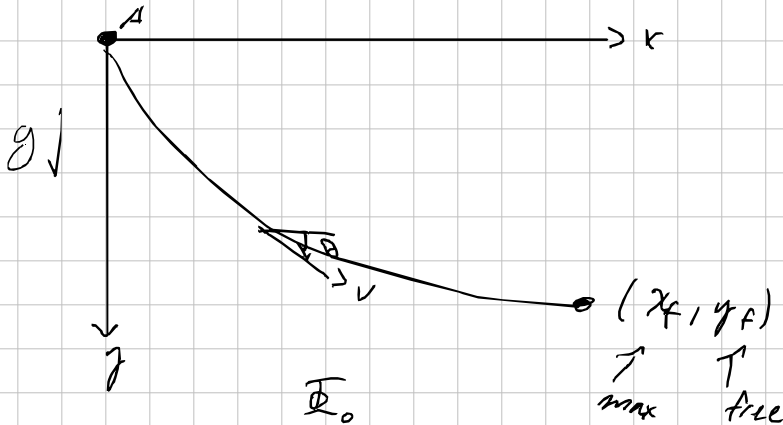
cancel

$$\rightarrow \dot{H}^* = \frac{\partial L}{\partial t} + \bar{\lambda}^T \frac{\partial \bar{f}}{\partial t}$$

if L and f are not explicit functions of time, then
 $\dot{H}^* = 0 \rightarrow H$ is constant.
integral of motion

almost always true.

Balistic Trajectory Problem



max final
x location
in given final
time, t_f .

$$J = \underbrace{[v_{0x} \ v_{0y}]}_{\vec{v}_0^T} \underbrace{\begin{bmatrix} x_0 \\ y_0 \end{bmatrix}}_{\vec{\psi}_0 = \vec{0}} - \underbrace{x_f}_{\phi_f = \Phi_f}$$

($L = 0$, no continuous cost)

$$0 = -mgy + \frac{1}{2}mv^2$$

$$v = \sqrt{2gy}$$

$$\dot{x} = \sqrt{2gy} \cdot \cos \theta$$

$$\dot{y} = \sqrt{2gy} \cdot \sin \theta$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \bar{x} \text{ state variables}$$

$$[\theta] = \bar{u} \text{ control variable}$$

Needing constraints for optimization
 Set up Lagrange Equations:

$$H = L + \bar{\lambda}^T \bar{f} = [\lambda_x \lambda_y] \begin{bmatrix} \sqrt{2gy} \cos \theta \\ \sqrt{2gy} \sin \theta \end{bmatrix}$$

$$H = \lambda_x (\sqrt{2gy} \cos \theta) + \lambda_y (\sqrt{2gy} \sin \theta)$$

E-L Eqs

$$I. \dot{\bar{\lambda}}^T = -\frac{\partial H}{\partial \bar{x}} = - \begin{bmatrix} 0 & \frac{g}{\sqrt{2gy}} (\lambda_x \cos \theta + \lambda_y \sin \theta) \end{bmatrix}$$

$$II. \bar{\lambda}(0) = -\frac{\partial \Phi_0}{\partial \bar{x}_0} = [-v_{0x}, -v_{0y}]$$

$$III. \bar{\lambda}(t_f) = \frac{\partial \Phi_f}{\partial \bar{x}_f} = [-1, 0]$$

$$IV. \frac{\partial H}{\partial u} = \frac{\partial H}{\partial \theta} = 0 = \underbrace{\sqrt{2gy}}_{\neq 0} (-\lambda_x \sin \theta + \lambda_y \cos \theta)$$

Solve everything.

Eqn IV.

$$\rightarrow \lambda_x \sin \theta = \lambda_y \cos \theta$$

$$\tan \theta = \pm \frac{\lambda_y}{\lambda_x} \quad (\text{expect } QI \text{ angle}) \quad \frac{+y}{+x}$$

$$\dot{\lambda}_x = 0 \quad \lambda_x(t_f) = -1 \rightarrow \therefore \lambda_x(t) = -1$$

$$\text{so, } \tan \theta = \frac{\cancel{0} \lambda_y}{\cancel{0} \lambda_x}$$

$$\tan \theta = -\frac{\lambda_y}{1} \rightarrow \text{"optimal control law"}$$

$L (= 0)$ and $\bar{F}$ (EDM) are not explicit fns of time (t) thro, $\dot{H} = 0$

$$\therefore H = \text{constant}$$

$$H = (-1)(\sqrt{2gy} \cos \theta) - \tan \theta (\sqrt{2gy} \sin \theta)$$

$$H = -\sqrt{2gy} \left[\cos \theta + \sin \theta \tan \theta \right]$$

$$\underbrace{\frac{\cos^2 \theta}{\cos \theta} + \frac{\sin^2 \theta}{\cos \theta}}$$

$$H = -\sqrt{2gy} \sec \theta = \text{constant.}$$

$$\dot{H} = -\frac{2gy}{2\sqrt{2gy}} \sec \theta - \sqrt{2gy} (\sec \theta \tan \theta) \dot{\theta}$$

$$\text{recall } y = \sqrt{2gy} \sin \theta$$

$$\dot{H} = -g \cancel{\tan \theta} + H \cancel{\tan \theta} \cdot \dot{\theta} = 0$$

$$\dot{\theta} = \frac{g}{H} = \text{constant.}$$

Ticantul Condiții

$$\tan \theta_f = \frac{-y(t_f)}{1} = \frac{0}{1} = 0 \rightarrow \therefore \theta_f = 0$$

$$\frac{d\theta}{dt} = \frac{g}{H}$$

$$d\theta = \frac{g}{H} dt$$

$$\int_{\theta(t)}^0 d\theta = \int_t^{t_f} \frac{g}{H} dt$$

$$-\theta(t) = \frac{g}{H} (t_f - t)$$

$$\theta(t) = -\frac{g}{H} (t_f - t)$$

$$\rightarrow H = -\sqrt{2gy_0} \sec \theta_0$$

$$H \cos \theta_0 = -\sqrt{2gy_0} = 0$$

$$\therefore \theta_0 = \frac{\pi}{2} \text{ rad.}$$

$$\theta(0) = \frac{\pi}{2} = -\frac{g}{H} (t_f - 0)$$

$$H = -\frac{2gt_f}{\pi}$$

$$\theta(t) = \frac{-g}{\left(-\frac{2gt_f}{\pi}\right)} (t_f - t) = \frac{\pi}{2} \left(1 - \frac{t}{t_f}\right)$$

$$\dot{y} = \sqrt{2gy} \sin \theta = \sqrt{2gy} \sin \left[\frac{\pi}{2} \left(1 - \frac{t}{t_f} \right) \right]$$

Separable 1st order ODE.

$$\frac{dy}{dt} = \dot{y} = \sqrt{2gy} \sin \left[\frac{\pi}{2} \left(1 - \frac{t}{t_f} \right) \right]$$

$$\int_0^{y(t)} \frac{dy}{\sqrt{y}} = \int_0^t \sqrt{2g} \cos \left(\frac{\pi t}{2t_f} \right) dt$$

$$\cancel{\sqrt{y(t)}} = \sqrt{2g} \left(\frac{2t_f}{\pi} \right) \sin \left(\frac{\pi t}{2t_f} \right)$$

$$y(t) = \frac{2g t_f^2}{\pi^2} \sin^2 \left(\frac{\pi t}{2t_f} \right)$$

using: $1 - \frac{\cos(2u)}{2} = \sin^2 u$

$$y(t) = \frac{g t_f^2}{\pi^2} \left[1 - \cos \left(\frac{\pi t}{t_f} \right) \right]$$

parameter eqn of a cycloid: $y(t) = r(1 - \cos(t))$

$$\dot{x} = \sqrt{2gy} \cos(\theta) = \sqrt{2g \left[\frac{2g t_f^2}{\pi^2} \sin^2 \left(\frac{\pi t}{2t_f} \right) \right]} \cos \left[\frac{\pi}{2} \left(1 - \frac{t}{t_f} \right) \right]$$

$$\frac{2g t_f^2}{\pi^2} \sin^2 \left(\frac{\pi t}{2t_f} \right) \quad \sin \left(\frac{\pi t}{2t_f} \right)$$

$$\dot{x} = \frac{2g t_f^2}{\pi^2} \sin^2 \left(\frac{\pi t}{2t_f} \right)$$

$$\int_0^{x(t)} dx = \frac{g t_f}{\pi} \int_0^{\pi} 1 - \cos\left(\frac{\pi t}{t_f}\right) dt$$

$$x(t) = \frac{g t_f}{\pi} \left[t - \frac{t_f}{\pi} \sin\left(\frac{\pi t}{t_f}\right) \right]$$

$$x(t) = \frac{g t_f^2}{\pi^2} \left[\frac{\pi t}{t_f} - \sin\left(\frac{\pi t}{t_f}\right) \right]$$

parametric form of cycloid: $x(t) = r(t - \sin t)$

1D Car

$$\min J = \frac{1}{2} \int_0^{t_f} u^2 dt \quad \text{"min energy"}$$

$$\text{subject to: } \begin{array}{lll} \dot{x} = v & x(0) = 0 & x(t_f) = x_f \\ \dot{v} = u & v(0) = 0 & v(t_f) = 0 \end{array}$$

State variables (Everything w/ Diff Eqn)

$$\rightarrow \bar{x} = \begin{bmatrix} x \\ v \end{bmatrix} \quad \bar{u} = u$$

$$H = L + \bar{\lambda}^T \bar{f} = \frac{1}{2} u^2 + [\lambda_x \lambda_v] \begin{bmatrix} v \\ u \end{bmatrix}$$

$$H = \frac{1}{2} u^2 + \lambda_x(v) + \lambda_v(u)$$

$$H = \frac{1}{2} u^2 + [\lambda_y \lambda_s] \begin{bmatrix} s \\ u - \frac{k_y}{m} \end{bmatrix}$$

$$\begin{aligned} \dot{y} &= s \\ \dot{s} &= \frac{u - k_y}{m} \end{aligned}$$

$$H = \frac{1}{2} u^2 + \lambda_y s + \lambda_s \left(\frac{u - k_y}{m} \right)$$

$$\dot{\lambda}^T = - \left[-\frac{k}{m} \lambda_s, \lambda_y \right]$$

$$u + \frac{\lambda_s}{m} = 0$$

$$u = -\frac{\lambda_s}{m}$$

$$\dot{\lambda}_y = -\frac{k}{m} \lambda_s \quad \dot{\lambda}_s = \lambda_y$$

$$\int_{\lambda_y(t)}^{v_1} d\lambda_y = \int_t^{t_f} -\frac{k}{m} \lambda_s dt$$

$$v_1 - \lambda_y(t) = -\frac{k}{m} \lambda_s (t_f - t)$$

$$\lambda_y(t) = \frac{k}{m} \lambda_s (t_f - t) + v_1$$

$$\int_{\lambda_s(t)}^{v_2} d\lambda_s = \int_t^{t_f} \frac{k}{m} \lambda_s (t_f - t) + v_1 dt$$

$$v_2 - \lambda_s(t) = \frac{k}{m} \lambda_s (t_f - t)^2 + v_1$$

$$\left[\lambda_s(t) = -\frac{k}{m} \lambda_s (t_f - t)^2 - v_1 + v_2 \right]$$

$$\Phi_f = \cancel{\psi_f}^0 + \bar{v}^T \bar{\psi}_f = [\bar{v}_1 \ \bar{v}_2] \begin{bmatrix} x(t_f) - x_f \\ v(t_f) \end{bmatrix}$$

E-L Equations

$$\dot{\bar{\lambda}}^T = -\frac{\partial H}{\partial \bar{x}} = -[0, \lambda_x]$$

$$\bar{\lambda}_f^T = \frac{\partial \Phi_f}{\partial \bar{x}_f} = [\bar{v}_1, \bar{v}_2]$$

$$\frac{\partial H}{\partial u} = \frac{\partial H}{\partial u} = 0 = u + \lambda_v$$

L and f are not explicit functions of time, t ,
there $\dot{H} = 0 \therefore H$ is constant.

$\Rightarrow u = -\lambda_v$ "optimal control law"

$$\dot{\lambda}_x = 0, \lambda_x(t_f) = \bar{v}_1 \rightarrow \therefore \lambda_x(t) = \bar{v}_1$$

$$\lambda_v = -\lambda_x = -\bar{v}_1, \lambda_v(t_f) = \bar{v}_2$$

$$\int_{\lambda_v(t)}^{\bar{v}_2} d\lambda_v = \int_t^{t_f} -\bar{v}_1 dt$$

$$\bar{v}_2 - \lambda_v(t) = -\bar{v}_1(t_f - t)$$

$$\lambda_v(t) = \bar{v}_2 + \bar{v}_1(t_f - t)$$

$$u(t) = -v_2 - v_1 (t_f - t)$$

$$\dot{v} = u = -v_2 - v_1 (t_f - t)$$

$$\int_0^{v(t)} dv = \int_0^t -v_2 - v_1 (t_f - t) dt$$

$$v(t) = -v_2 t + \frac{1}{2} v_1 (t_f - t)^2 - \frac{1}{2} v_1 t_f^2$$

$$v(t) = \dot{x} = -v_2 t + \frac{1}{2} v_1 (t_f - t)^2 - \frac{1}{2} v_1 t_f^2$$

$$\int_0^{x(t)} dx = \int_0^t -v_2 t + \frac{1}{2} v_1 (t_f - t)^2 - \frac{1}{2} v_1 t_f^2 dt$$

$$x(t) = -\frac{1}{2} v_2 t^2 - \frac{1}{6} v_1 (t_f - t)^3 - \frac{1}{2} v_1 t_f^2 t + \int_0^t$$

$$x(t) = -\frac{1}{2} v_2 t^2 - \frac{1}{6} v_1 (t_f - t)^3 - \frac{1}{2} v_1 t_f^2 t + \frac{1}{6} v_1 t_f^3$$

$$x(t_f) = x_f = -\frac{1}{2} v_2 t_f^2 - \frac{1}{6} v_1 t_f^3 + \frac{1}{6} v_1 t_f^3$$

$$x(t_f) = \left(-\frac{t_f^2}{2}\right) v_1 + \left(-\frac{t_f^3}{3}\right) v_2$$

$$v(t_f) = 0 = -v_2 t_f - \frac{1}{2} v_1 t_f^2$$

$$v(t_f) = 0 = \left(-\frac{t_f^2}{2}\right) v_1 - (t_f) v_2$$

$$\begin{bmatrix} -\frac{+f^3}{3} & -\frac{+f^2}{2} \\ -\frac{1}{2}f^2 & -f \end{bmatrix}^{-1} \begin{bmatrix} x_f \\ 0 \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} : \frac{1}{\frac{+f^4}{3} - \frac{+f^4}{4}} \begin{bmatrix} -f & f_{f/2}^2 \\ \frac{+f^2}{2} & -\frac{+f^3}{3} \end{bmatrix} \begin{bmatrix} x_f \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \frac{12}{+f^4} \begin{bmatrix} -f x_f \\ \frac{+f^2}{2} x_f \end{bmatrix}$$

$$v_1 = -\frac{12 x_f}{+f^3}$$

$$v_2 = \frac{6 x_f}{+f^2}$$

Now replace in EoB

$$u(t) = -\frac{6x_f}{t_f^2} + \left(\frac{12x_f}{t_f^3} \right) (t_f - t)$$

$$u(t) = \frac{6x_f}{t_f^2} - \frac{12x_f}{t_f^3} \cdot t = \frac{6x_f}{t_f^2} \left(1 - 2\frac{t}{t_f} \right)$$

$$v(t) = -\frac{6x_f}{t_f^2} \cdot t + \frac{1}{2} \left(\frac{-12x_f}{t_f^3} \right) (t_f - t)^2 - \frac{1}{2} \cdot \frac{-12x_f}{t_f^3} \cdot t_f^2$$

$t_f^2 - 2t_f t + t^2$

$$v(t) = \frac{6x_f}{t_f^2} \cdot t - \frac{6x_f}{t_f^2} t^2 = \frac{6x_f}{t_f^2} \left[\left(\frac{t}{t_f} \right) - \left(\frac{t}{t_f} \right)^2 \right]$$

$$x(t) = 6x_f \left[\frac{t^2}{2t_f} - \frac{t^3}{3t_f} \right]$$

$$x(t) = x_f \left[3\left(\frac{t}{t_f}\right)^2 - 2\left(\frac{t}{t_f}\right)^3 \right]$$

could plug into

$$\hookrightarrow H = \frac{1}{2} u^2 + \lambda_u(v) + \lambda_v(u)$$

Linear Quadratic Regulator

$$\min J = \frac{1}{2} \bar{x}_f^T [S_f] \bar{x}_f + \frac{1}{2} \int_0^T \bar{x}^T [Q] \bar{x} + \bar{u}^T [R] \bar{u} dt$$

$$\text{subject to: } \dot{\bar{x}} = [A] \bar{x} + [B] \bar{u}$$

$$\bar{\Psi} = \bar{0} = [u_f] \bar{x}_f - \bar{c}$$

Linear Equation of Motion
Quadratic Cost

Prescribed, control $\bar{x}$

$\bar{x}$: states ($n \times 1$)

$\bar{u}$: controls ($m \times 1$)

$\bar{\Psi}$: constraints ($q \times 1$)

$$J = \frac{1}{2} \underbrace{\bar{x}_f^T}_{1 \times n} \underbrace{[S_f]}_{n \times n} \underbrace{\bar{x}_f}_{n \times 1} + \frac{1}{2} \int_0^T \underbrace{\bar{x}^T}_{(1 \times n)} \underbrace{[Q]}_{(n \times n)} \underbrace{\bar{x}}_{(n \times 1)} + \underbrace{\bar{u}^T}_{1 \times m} \underbrace{[R]}_{m \times m} \underbrace{\bar{u}}_{m \times 1} dt$$

$[S_f], [Q], [R]$ = symmetric weighting matrices

$$\underbrace{\dot{\bar{x}}}_{n \times 1} = \underbrace{[A]}_{n \times n} \underbrace{\bar{x}}_{n \times 1} + \underbrace{[B]}_{n \times m} \underbrace{\bar{u}}_{m \times 1} \quad \bar{\Psi} = \bar{0} = \underbrace{[u_f]}_{q \times n} \underbrace{\bar{x}_f}_{n \times 1} - \underbrace{\bar{c}}_{q \times 1}$$

$$J = \underbrace{\frac{1}{2} \bar{x}_f^T [S_f] \bar{x}_f + \bar{v}^T \psi}_{\Phi_f} + \frac{1}{2} \int_0^t \bar{x}^T [Q] \bar{x} + \bar{u}^T [R] \bar{u} dt$$

$$H = \frac{1}{2} (\bar{x}^T Q \bar{x} + \bar{u}^T R \bar{u}) + \bar{\lambda}^T (A \bar{x} + B \bar{u})$$

Euler Lagrange Eqns:

$$\dot{\bar{\lambda}}^T = - \frac{\partial H}{\partial \bar{x}} = - (\bar{x}^T Q + \bar{\lambda}^T A)$$

$$\bar{\lambda}^T(t_f) = \frac{\partial \Phi_f}{\partial \bar{x}_f} = \bar{x}_f^T S_f + \bar{v}^T \psi_f$$

$$\frac{\partial H}{\partial \bar{u}} = 0 = \bar{u}^T R + \bar{\lambda}^T B \quad \text{solve first}$$

Solve:

$$\begin{aligned} \bar{u}^T R &= -\bar{\lambda}^T B \\ \Rightarrow R \bar{u} &= -B^T \bar{\lambda} \end{aligned}$$

$$\bar{u} = -R^{-1} B^T \bar{\lambda} \quad \text{optimal control law}$$

Then plug into differential eqns (continuous)

$$\dot{\bar{x}} = A \bar{x} - B R^{-1} B^T \bar{\lambda} \quad \bar{x}(0) = \bar{x}_0$$

$$\dot{\bar{\lambda}} = -Q \bar{x} - A^T \bar{\lambda} \quad \bar{\lambda}(t_f) = S_f \bar{x}_f + \psi_f^T \bar{v}$$

$$\frac{u - km}{\delta}$$

-

$$\frac{k_s}{m} \lambda_s - \lambda_y$$

$$\frac{-b_s}{m}$$

$$\text{let } \lambda(t) = [S(t)] \bar{x}(t) + [V(t)] \bar{v}$$

$$\text{then } S(t_f) = S_f ; \quad V(t_f) = U_f^T$$

$$\rightarrow \dot{\lambda} = \dot{S} \bar{x} + S \dot{\bar{x}} + \dot{V} \bar{v} = -Q \bar{x} - A^T \underbrace{(S \bar{x} + V \bar{v})}_{\lambda}$$

$$\dot{\lambda} = \dot{S} \bar{x} + S[A \bar{x} - B R^{-1} B^T (S \bar{x} + V \bar{v})] + \dot{V} \bar{v} = -Q \bar{x} - A^T (S \bar{x} + V \bar{v})$$

$$\text{x terms: } \underbrace{\dot{S} + S A + A^T S - S B R^{-1} B^T S + Q = 0}_{S(t_f) = S_f} \quad \text{MATRIX RICCATI EQUATION}$$

yields $S(t)$

for problems when $t_f \rightarrow \infty$ (or is very large)
essentially if you have a single controller,
then $\dot{S} = 0$ (S constant) (no slope)

$$\underbrace{S A + A^T S - S B R^{-1} B^T S + Q = 0}_{\text{Algebraic Riccati Equation.}} \quad \text{results in constant } S$$

$$[S] = \text{icare}(A, B, Q, R)$$

'implicit continuous algebraic Riccati' eqn.

$$\bar{v} \text{ terms: } -SBR^{-1}B^T V + \dot{V} = -A^T V$$

$$\rightarrow \dot{V} = (SBR^{-1}B^T - A^T)V \quad V(t_f) = U_f^T$$

$\begin{matrix} n \times n & (n \times m) & (m \times m) & (m \times m) & (m \times m) \end{matrix} \quad n \times g$

$V(t_f) = U_f^T$
 $n \times g$

Results in $V(t)$

$$\text{but } \bar{u} = -A^{-1}B^T \bar{\lambda} = -K^{-1}B^T(S\bar{x} + V\bar{v})$$

5241 need $\bar{v}$

$$\bar{\varphi} = U_f \bar{x}_f - \bar{z} = \bar{0}$$

$\begin{matrix} \text{constant} & & q \times n & n \times 1 & & q \times q & q \times 1 \end{matrix}$

$$\bar{z} = U_f \bar{x}_f = [U(t)]x(t) + [P(t)]\bar{v}$$

$\underbrace{\quad\quad\quad}_{\text{Let}} \quad u(t_f) = U_f, \quad [P(t_f)] = 0 (q \times q)$

$$\bar{0} = \dot{u} \bar{x} + u \dot{\bar{x}} + \dot{P} \bar{v}$$

$$\bar{0} = \dot{u} \bar{x} + u [A \bar{x} - BR^{-1}B^T(S\bar{x} + V\bar{v})] + \dot{P} \bar{v}$$

$$\bar{x} \text{ terms: } \bar{0} = \dot{u} + uA - uBR^{-1}B^TS$$

$$\dot{u} = u(BR^{-1}B^TS - A) \quad u(t_f) = U_f$$

$$\bar{v} \text{ terms: } \bar{0} = -uBR^{-1}B^TV + \dot{P}$$

$$\dot{P} = V^T B R^{-1} B^T V \quad P(t_f) = \bar{0} \quad q \times q$$

yield $P(t)$

$$z = V_f^T x_f = V_f^T \bar{x} + P \bar{v}$$

$$\bar{v} = P^{-1} (V_f^T \bar{x}_f - V_f^T \bar{x})$$

DNE @ t_f

thus:

$$\bar{u} = -K^T B^T [S \bar{x} + V P^{-1} (V_f^T x_f - V_f^T \bar{x})]$$

$$\dot{\bar{x}} = [A] \bar{x} + [B] \bar{u}$$

3.2-8

$$J = S_{ff} (t_f)^2 + \frac{1}{2} \int_0^{t_f} (x_1^2 + x_2^2 + u^2) dt$$

subject to $\dot{x}_1 = x_2 + u$
 $\dot{x}_2 = x_1 - x_2$, $x_2(t_f) = 1$

Let $x_1(0) = x_2(0) = 0$

Algebraic Riccati Eqn (ARE):

$$SA + A^T S - SBR^{-1}B^T S + Q = 0$$

$$\dot{\bar{x}} = \begin{bmatrix} 0 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_B u$$

$$J = \int_0^\infty \underbrace{\begin{bmatrix} x_1 & x_2 \end{bmatrix}}_{\bar{x}^T} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_Q \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + u \underbrace{(1)}_R u dt$$

$$S = \begin{bmatrix} \sqrt{3} & 1 \\ 1 & 1 \end{bmatrix}$$

$$\bar{u} = -R^{-1}B^T \bar{S} \bar{x} = -R^{-1}B^T (S\bar{x}) = -(1)^{-1} \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \sqrt{3} & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

new variables: $x_1' = x_1$

$x_2' = x_2 - 1$ ← desired x_2 @ t_f

$$\text{then } \dot{x}_1' = \dot{x}_1 = (x_2' + 1) + u$$

$$\rightarrow \dot{x}_2' = \dot{x}_2 = x_1' - (x_2' + 1)$$

$$\bar{u} = -(1)^{-1} [1 \ 0] \begin{bmatrix} \sqrt{3} & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1' \\ x_2' \end{bmatrix} = -[\sqrt{3} \ 1] \begin{bmatrix} x_1' \\ x_2' \end{bmatrix}$$

$$\bar{u} = -\sqrt{3}x_1' - x_2'$$

$$u = -\sqrt{3}x_1 - x_2 + \sqrt{3} \quad \left\{ \begin{array}{l} \text{recall } \dot{x}_1 = x_2 + u \\ \text{assume } \dot{x}_1(t_f) = \dot{x}_2(t_f) = 0 \end{array} \right.$$

$$x_1 = x_2 = 1$$

$$u = -1 = -\sqrt{3}(1) - (1) + c$$

$$c = \sqrt{3} - 2$$

$$\underline{\|x(t_f)\|^2}$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \underbrace{\begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix}}_{S_f} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\dot{S} + SA + A^T S - SBR^{-1}B^T S + Q = 0 \quad S(t_f) = S_f$$

$$\dot{V} = (SBR^{-1}B^T - A^T)V \quad V(t_f) = u_f^T$$

$$\bar{\Psi} = \underbrace{[u_f]}_{2 \times 1} \underbrace{\bar{x}_f}_{2 \times 1} - \underbrace{\bar{c}}_{1 \times 1} = \underbrace{\bar{0}}_{1 \times 1} \quad \rightarrow x_2(t_f) - 1 = 0$$

$$\bar{\Psi} = \underbrace{[0 \ 1]}_{u_f} \begin{bmatrix} x_{1f} \\ x_{2f} \end{bmatrix} - 1 = 0$$

$$u_f^T = u_f = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$S(t_f) = S_f = \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\dot{V} = \underbrace{(SBR^{-1}B^T - A^T)}_{\substack{(2 \times 2)(2 \times 1)(1 \times 1)(1 \times 2) \\ 2 \times 2}} V \quad \begin{matrix} 2 \times 1 \\ 2 \times 2 \end{matrix}$$

$$\bar{p} = \underbrace{V^T}_{1 \times 2} \underbrace{B}_{2 \times 1} \underbrace{R^{-1}}_{1 \times 1} \underbrace{B^T}_{1 \times 2} \underbrace{V}_{2 \times 1} \quad p(t_f) = 0 \quad (1 \times 1)$$

$$\bar{u} = -R^{-1}R^T[S\bar{x} + V\rho^{-1}(\underbrace{\bar{V}_f^T}_{\substack{w \\ \Sigma^0 \quad 1}}\bar{x}_f - \underbrace{V^T\bar{x}}_{\substack{\uparrow \\ [u \\ 1]}})]$$

$$\dot{\bar{x}} = A\bar{x} + B\bar{u}$$

$$x_1' = x_1 - 1$$

$$x_2' = x_2 + 1 \quad \longrightarrow \quad \text{managing to derive } x_2(t) = 1 \text{ constant}$$

$$u' = u + 1$$

$$\dot{x}_1' = \dot{x}_1 = x_2 + u = (x_2' + 1) + (u' - 1) = x_2' + u'$$

$$\dot{x}_2' = \dot{x}_2 = x_1 - x_2 = (x_1' + 1) - (x_2' + 1) = x_1' - x_2'$$

$$J = \frac{1}{2} \int_0^{\infty} \bar{x}^T [Q] \bar{x} + \bar{u}^T [R] \bar{u} \, dt$$

Optimize Nozzle core shape to minimize drag.

$$F_D = \rho_{\infty} \int C_p(\theta) dA$$

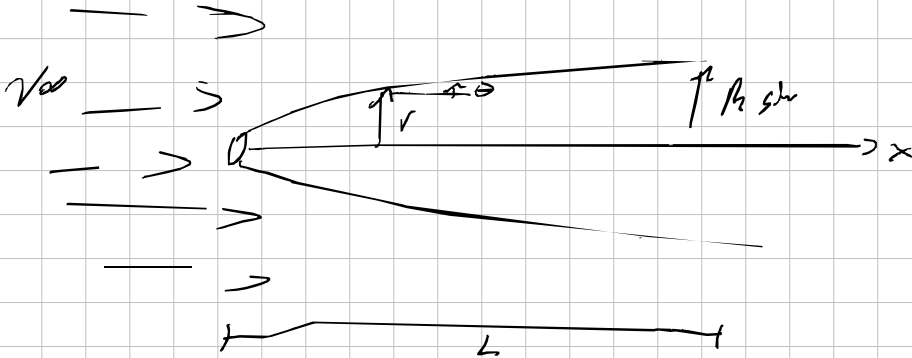
$$\rho_{\infty} = \frac{1}{2} \rho_{\infty} V_{\infty}^2$$

ρ_{∞} free stream density

V_{∞} velocity of free stream air

C_p pressure coefficient

$$C_p = \begin{cases} 2\sin^2(\theta) & \theta \geq 0 \\ 0 & \theta < 0 \end{cases}$$



$$F_D = \rho_{\infty} \int C_p(\theta) 2\pi r dr = \rho_{\infty} \int_{r_0}^R (2\sin^2\theta) (2\pi r) dr$$

L, R given

$$F_D = 4\pi\eta\omega \int_{r_0}^R r \sin^2 \theta \, dr$$

For Sent tip, $\omega = 2$

$$\text{so } F_{D_0} = 4\pi\eta\omega \int_0^{r_0} r \, dr = 4\pi\eta\omega \left(\frac{r_0^2}{2} \right)$$

$$J = \frac{F_D}{4\pi\eta\omega} = \frac{r_0^2}{2} + \underbrace{\int_{r_0}^R r \sin^2 \theta \, dr}_{u \, dx}$$

Must satisfy Diff Eq



$$\frac{dr}{dx} = \tan \theta = u = \frac{\sin \theta}{\cos \theta} \quad \text{recall } 1 + \tan^2 \theta = \sec^2 \theta$$

$$u^2 = \frac{\sin^2 \theta}{\cos^2 \theta} = \sin^2 \theta (1 + u^2)$$

$$J = \frac{r_0^2}{2} + \int_0^L \frac{r u^3}{1 + u^2} \, dx \quad \text{subject to: } \frac{dr}{dx} = u, \quad r(0) = \text{free}, \quad r(L) = R$$

Do have a terminal constraint

$$J = \underbrace{\frac{1}{2} v(0)^2}_{\Phi_0} + \underbrace{v(r(L) - R)}_{\text{adding in terminal constraint}} + \int_0^L \frac{ru^3}{1+u^2} dx$$

$$H = \frac{ru^3}{1+u^2} + \lambda(u)$$

$E - L$:

$$\dot{\lambda} = -\frac{\partial H}{\partial v} = -\frac{u^3}{1+u^2} \quad \text{where } \lambda(0) = -\frac{\partial \Phi_0}{\partial v_0} = -v_0$$

$$\lambda(L) = \frac{\partial \Phi_f}{\partial r_f} = v$$

$$\frac{\partial H}{\partial u} = 0 = \frac{3ru^2}{1+u^2} - \frac{ru^3(2u)}{(1+u^2)^2} + \lambda$$

H not explicit for x

λ and f not explicit functions of
then independent variables (x)

$\rightarrow H$ constant

Solve $\frac{\partial \mathcal{L}}{\partial u}$ for λ instead of for optimal control

$$\lambda = \frac{2ru^4 - 3ru^2(1+u^2)}{(1+u^2)^2} = \frac{-3ru^2 - ru^4}{(1+u^2)^2}$$

$$\lambda = \frac{-ru^2(3+u^2)}{(1+u^2)^2}$$

@ initial $x=0$, $\lambda(0) = -r_0$

$$-r_0 = \frac{-r_0 u_0^2 (3+u_0^2)}{(1+u_0^2)^2}$$

$$\begin{array}{c} 0 \\ \neq 0 \end{array} = r_0 \underbrace{\left[1 - \frac{u_0^2 (3+u_0^2)}{(1+u_0^2)^2} \right]}_{=0}$$

$$\frac{u_0^2 (3+u_0^2)}{(1+u_0^2)^2} = 1$$

$$3u_0^2 + \cancel{u_0^4} = 1 + 2u_0^2 + \cancel{u_0^4}$$

$$u_0^2 = 1$$

$$u_0 = \pm 1 = \tan(\theta_0)$$

$$\rightarrow \theta_0 = 45^\circ$$

$$H_0 = H(x) = \frac{r_0 (y_0')^3}{(1+u^2)^2} + (-r_0) (y_0')' = \frac{r_0}{2} - r_0 = -\frac{r_0}{2}$$

for all x .

$$H = -\frac{r_0}{2} = \frac{ru^3}{1+u^2} + \frac{-ru^2(3+u^2)}{(1+u^2)^2} \cdot u$$

$$H = ru^3 \left(\frac{(1+u^2) - (3+u^2)}{(1+u^2)^2} \right)$$

$$H = -\frac{r_0}{2} = \frac{2ru^3}{(1+u^2)^2}$$

$$r = \frac{r_0}{4} \cdot \frac{(1+u^2)^2}{u^3} = \frac{r_0}{4} \left(\frac{1}{u^3} + \frac{2}{u} + u \right)$$

EOM (~~Dx/Eg~~) $\rightarrow$

$$\frac{dr}{dx} = u$$

$$\frac{dr}{du} = dx$$

$$\frac{d}{du}$$

$$\frac{dr}{du} = \frac{r_0}{4} \left(1 - \frac{3}{u^4} - \frac{2}{u^2} \right)$$

$$dr = \frac{r_0}{4} \left(1 - \frac{3}{u^4} - \frac{2}{u^2} \right) du$$

$$\int_{u_0=1}^{u(x)} \frac{r_0}{4} \left(\frac{1}{u} - \frac{3}{u^5} - \frac{2}{u^3} \right) du = \int_0^x dx$$

$$\frac{r_0}{4} \left(\ln(u) + \frac{3}{4} \frac{1}{u^4} + \frac{1}{u^2} \right) \Big|_1^{u(x)} = x$$

$$\frac{r_0}{4} \left[\ln(u(x)) + \frac{3}{4u^4} + \frac{1}{u^2} - \frac{3}{4} - 1 \right] = x$$

$$\boxed{x = \frac{r_0}{4} \left[\ln(u) + \frac{3}{4u^4} + \frac{1}{u^2} - \frac{7}{4} \right]}$$

given x , solve for u (numerically)
 then w/ u : solve for r

Need $r_0 \rightarrow$ use $x=L$

$$L = \frac{r_0}{4} \left(\ln(u_L) + \frac{3}{4u_L^4} + \frac{1}{u_L^2} - \frac{7}{4} \right)$$

$$\left\{ \frac{r_0}{4} \left(\frac{1}{u_L^3} + \frac{2}{u_L} + u_L \right) = r(L) = R \right.$$

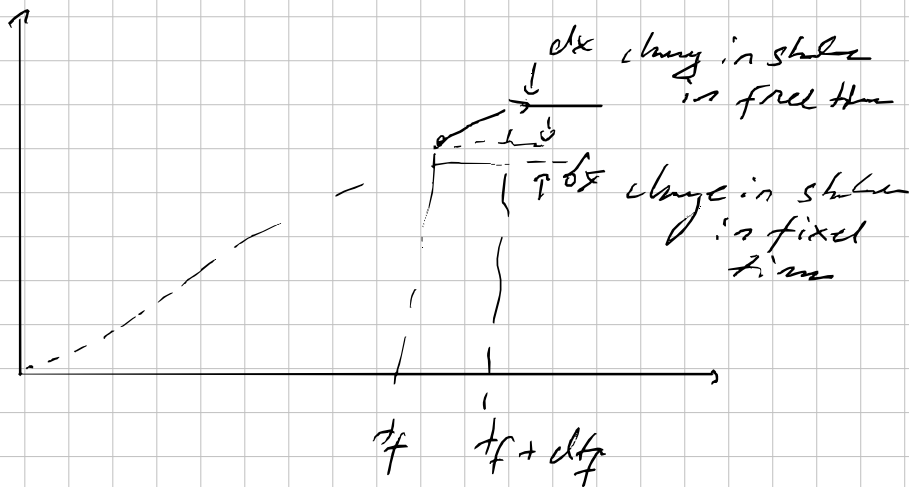
solve for u_L and r_0

Assume

$$J = \Phi_f(\bar{x}_f, t_f) + \int_{t_0}^{t_f} L(\bar{x}, \bar{u}, t) dt$$

$$(no \Phi_0(\bar{x}_0, t_0)) \text{ sub } \bar{x} = \bar{f}(\bar{x}, \bar{u}, t)$$

$$d\bar{x} = \delta\bar{x} + \dot{\bar{x}} dt_f$$



$$H(\bar{x}, \bar{u}, t) = L + \bar{\lambda}^T \bar{f}$$

$$J = \Phi_f(\bar{x}_f, t_f) + \int_{t_0}^{t_f} L + \bar{\lambda}^T (\bar{f} - \dot{\bar{x}}) dt$$

$$J = \Phi_f(\bar{x}_f, t_f) + \int_{t_0}^{t_f} H(\bar{x}, \bar{u}, t) - \bar{\lambda}^T \dot{\bar{x}} dt$$

Lagrange Integral rule

$$dJ = \frac{\partial \Phi_f}{\partial \bar{x}_f} d\bar{x}_f + \frac{\partial \Phi_f}{\partial t_f} dt_f + \left(H(\bar{x}, \bar{u}, t) - \bar{\lambda}^T \dot{\bar{x}} \right) dt \Big|_{t_0}^{t_f}$$

$$+ \int_{t_0}^{t_f} \left(\frac{\partial H}{\partial \bar{x}} \delta \bar{x} + \frac{\partial H}{\partial \bar{u}} \delta \bar{u} - \bar{\lambda}^T \delta \dot{\bar{x}} \right) dt$$

Integrate by parts

$$- \int_{t_0}^{t_f} \bar{\lambda}^T \delta \dot{\bar{x}} dt = - \bar{\lambda}^T \delta \bar{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \dot{\bar{\lambda}}^T \delta \bar{x} dt$$

$$u = -\bar{\lambda}^T \quad v = \delta \bar{x}$$

$$du = -\dot{\bar{\lambda}}^T dt \quad dv = \delta \dot{\bar{x}} dt$$

$$\Rightarrow \delta \dot{\bar{x}} = d\bar{x} - \dot{\bar{x}} dt$$

$$- \int_{t_0}^{t_f} \bar{\lambda}^T \delta \dot{\bar{x}} dt = - \bar{\lambda}^T (d\bar{x} - \dot{\bar{x}} dt) \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \dot{\bar{\lambda}}^T \delta \bar{x} dt$$

$$dJ = \left(\frac{\partial \Phi_f}{\partial \bar{x}_f} - \bar{\lambda}^T \right) d\bar{x}_f + \left(\frac{\partial \Phi_f}{\partial t_f} + H - \cancel{\bar{\lambda}_f^T \dot{\bar{x}}_f} + \cancel{\bar{\lambda}_f^T \dot{\bar{x}}_f} \right) dt_f$$

$$+ (\bar{\lambda}_0^T) d\bar{x}_0 + (-H + \cancel{\bar{\lambda}_0^T \dot{\bar{x}}_0} - \cancel{\bar{\lambda}_0^T \dot{\bar{x}}_0}) dt_0$$

$$+ \int_{t_0}^{t_f} \left(\frac{\partial H}{\partial \bar{x}} + \dot{\bar{\lambda}}^T \right) \delta \bar{x} + \left(\frac{\partial H}{\partial \bar{u}} \delta \bar{u} \right) dt = 0$$

Euler Lagrange Equations for free time

$$\dot{\lambda}^T = -\frac{\partial H}{\partial x}$$

$$\dot{\lambda}^T(t_f) = \frac{\partial \Phi_f}{\partial x_f}$$

either $dx_i(0) = 0$
fixed initial state

or $\lambda_i(0) = 0$

free initial state

$$\frac{\partial H}{\partial u} = 0$$

if: $dt_0 \neq 0$ (free initial time)

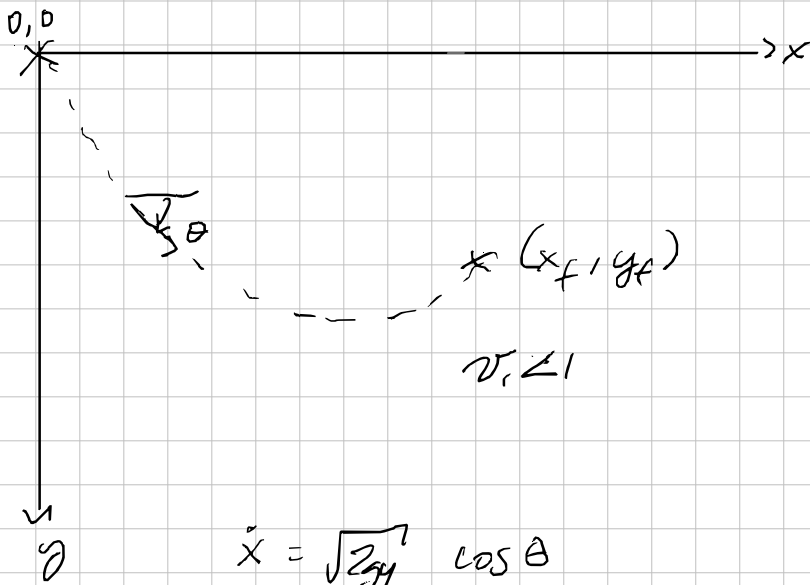
then: $H(0) = 0$

if: $dt_f \neq 0$ (free final time) (free end time)

$$H(t_f) = -\frac{\partial \Phi_f}{\partial t_f}$$



Transversality Condition



min time problem:

$$J = \varphi_f(\bar{x}(t_f), t_f) + \int_0^{t_f} L(\bar{x}, \bar{u}, t) dt$$

2 ways
to write

$$J = t_f, \quad \varphi_f = t_f, \quad L = 0$$

$$\text{or } J = \int_0^{t_f} 1 dt, \quad \varphi_f = 0, \quad L = 1$$

$$H = L + \bar{\lambda}^T \bar{f}$$

$$\text{transversality cond: } H(t_f) = -\frac{d\bar{\Phi}_f}{dt} = -1 = \bar{\lambda}_f^T \bar{f}_f$$

$$\bar{p}_f = \bar{\Phi}_f \text{ (no } x_f, y_f \text{ + comp)}$$

$$J: t_f \rightarrow H(t_f) = -1$$

$$J = \int_0^{t_f} 1 dt \rightarrow \boxed{H(t_f) = 0} \quad 1 + \lambda_f^T \bar{f}_f \dots \bar{\lambda}_f^T \bar{f} = -1$$

usually choose

$$J = v_1 |x(t_f) - x_f| + v_2 |y(t_f) - y_f| + \int_0^{t_f} 1 dt$$

$$H = 1 + \lambda_x (\sqrt{2g} \cos \theta) + \lambda_y (\sqrt{2g} \sin \theta)$$

E-L eqns:

$$\dot{\lambda}^T = -\frac{\partial H}{\partial x} = - \left[0, \frac{2g}{\sqrt{2g}} (\lambda_x \cos \theta + \lambda_y \sin \theta) \right]$$

$$\lambda^T(t_f) = \frac{\partial \Phi}{\partial x(t_f)} = [v_1, v_2]$$

$$\frac{\partial H}{\partial u} = 0 = \frac{\partial H}{\partial \theta} = \underbrace{\sqrt{2g}}_{\neq 0} \underbrace{(-\lambda_x \sin \theta + \lambda_y \cos \theta)}_{=0}$$

$$H(t_f) = \frac{\partial \Phi}{\partial t_f} = 0$$

usually, L & f are not explicit fns of time,
so $H = 0 \therefore H$ is constant.

$\rightarrow \therefore H$ for all time ($H(t) = 0$).

$$\frac{dH}{d\theta} = 0$$

$$l_x \sin \theta = l_y \cos \theta$$

$$\tan \theta = \frac{\pm l_y}{\pm l_x}$$

$$\dot{l}_x = 0, \therefore l_x(t) = v_1$$

$$\rightarrow \tan \theta = \frac{\pm l_y}{\pm v_1} \quad \left\{ \begin{array}{l} \text{choose } +l_y \end{array} \right.$$

$$H = 1 + \sqrt{2g y} (v_1 \cos \theta + (v_1 \tan \theta) \sin \theta) = 0$$

$$v_1 \sqrt{2g y} \underbrace{\left(\cos \theta + \frac{\sin^2 \theta}{\cos \theta} \right)}_{\sec \theta} = -1$$

$$v_1 \sqrt{2g y} = -\cos \theta$$

$$@ t = 0, \rightarrow y = 0$$

$$\Rightarrow 0 = -\cos(\theta)$$

$$\theta_0 = \pi/2 \quad (\text{straight down})$$

$$H = 1 + v_1 \sqrt{z y} \sec \theta = 0$$

$\frac{d}{dt} \downarrow$

$$\dot{H} = 0 = \frac{dy}{dt} \cdot \frac{1}{\sqrt{z y}} + v_1 \sec \theta + v_1 \sqrt{z y} \cdot \sec \theta \tan \theta \cdot \dot{\theta}$$

$$0 = g v_1 \tan \theta + v_1 \sqrt{z y} \sec \theta \tan \theta \dot{\theta}$$

-1

$$0 = g v_1 - \dot{\theta}$$

$$\dot{\theta} = g v_1 = \text{constant.}$$

$$\int_{\pi/2}^{\theta(t)} d\theta = \int_0^t v_1 g dt$$

$$\theta(t) - \pi/2 = v_1 g t$$

$$\boxed{\theta(t) = v_1 g t + \pi/2}$$

$$\dot{y} = \sqrt{z y} \sin \left(\frac{\pi}{2} + v_1 g t \right)$$

$\cos(v_1 g t)$

$$\int_0^{y(t)} \frac{dy}{\sqrt{y}} = \sqrt{z} \int_0^t \cos(v_1 g t) dt$$

$$2\sqrt{y(t)}^2 = \sqrt{2g} \cdot \frac{1}{v_1 g} \sin(v_1 g t)$$

$$y(t) = \frac{\sin^2(v_1 g t)}{2v_1^2 g}$$

↙

$$y(t) = \frac{1}{4v_1^2 g} [1 - \cos(2v_1 g t)]$$

invariant cycloid form

$$\dot{x} = \sqrt{2gy} \cos \theta = \sqrt{2g} \cdot \sqrt{\frac{1}{2v_1^2 g}} \cdot \sin(v_1 g t) \underbrace{\cos\left(\frac{\pi}{2} + v_1 g t\right)}_{-\sin(u)}$$

$$\dot{x} = -\frac{1}{v_1} \sin^2(v_1 g t) = -\frac{1}{2v_1} (1 - \cos(2v_1 g t))$$

$$\int_0^{x(t)} dx = \frac{1}{2v_1} \int_0^t -1 + \cos(2v_1 g t) dt$$

$$x(t) = \frac{1}{2v_1} \left[-t + \frac{1}{2v_1 g} \sin(2v_1 g t) \right]_0^t$$

$$x(t) = \frac{1}{4v_1^2 g} [-2v_1 g t + \sin(2v_1 g t)]$$

invariant cycloid form.

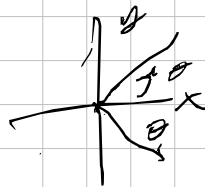
$$\left\{ \begin{aligned} x(t_f) = x_f &= \frac{1}{4\sigma_1^2 g} (-2\sigma_1 g t_f + \sin(2\sigma_1 g t_f)) \\ y(t_f) = y_f &= \frac{1}{4\sigma_1^2 g} (1 - \cos(2\sigma_1 g t_f)) \end{aligned} \right.$$

2 nonlinear transcendental equations
 $\rightarrow$ must solve numerically for
 σ_1, t_f .

optimal control law

$$\tan \theta = \frac{\pm \lambda y}{\pm \sigma_1}$$

$x_f > 0, \sigma_1 < 0, \rightarrow \theta$ is QI or QIV



$$\tan \theta = \frac{y}{+x}$$

↖
write as
pos.

must choose: $-\frac{\lambda y}{\sigma_1}$

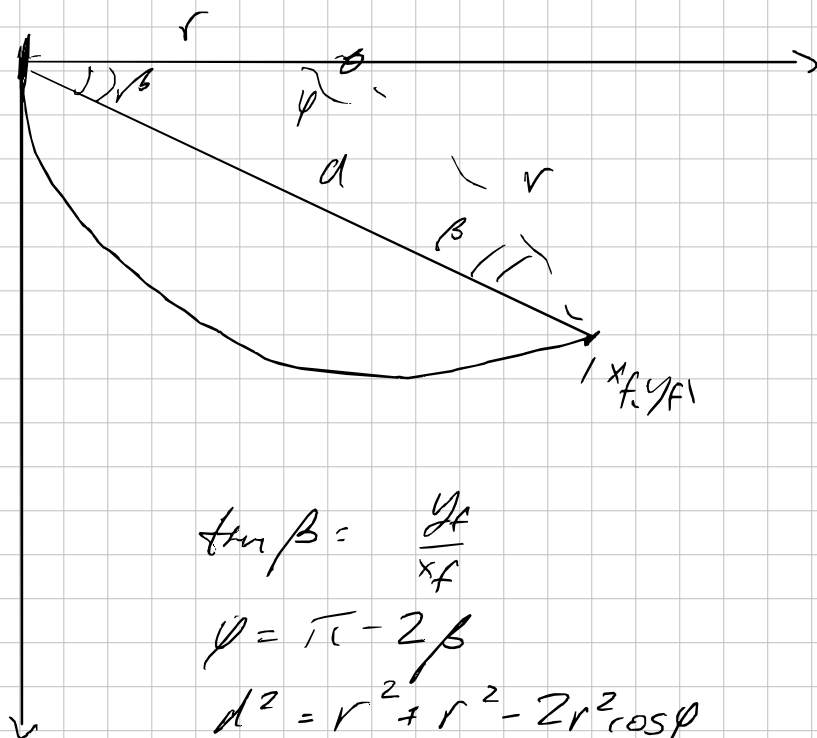
$-(-25)$

$x_f < 0, \sigma_1 > 0, \theta$ is QII or QIII



$$\tan \theta = \frac{y}{-x} \rightarrow \text{must choose } -\frac{\lambda y}{-\sigma_1}$$

as well.



$$\tan \beta = \frac{y_f}{x_f}$$

$$\varphi = \pi - 2\beta$$

$$d^2 = r^2 + r^2 - 2r^2 \cos \varphi$$

$$d^2 = 2r^2 (1 - \cos(\pi - 2\beta))$$

$$d^2 = 2r^2 (1 + \cos(2\beta))$$

$$d^2 = 2r^2 (1 + \cos^2 \beta - \sin^2 \beta)$$

$$d^2 = 4r^2 \cos^2 \beta$$

$$d = 2r \cos \beta = 2r \frac{x_f}{d}$$

$$\frac{d^2}{2x_f} = r$$

$$\tau = \frac{t}{t_f}$$

$$\frac{d\tau}{dt} = \frac{1}{t_f}$$

$$\frac{d\bar{x}}{dt} = f(\bar{x}, \bar{u}, t)$$

$$\frac{d\bar{x}}{d\tau} = \frac{d\bar{x}}{dt} \cdot \underbrace{\frac{dt}{d\tau}}_{t_f} = f(\bar{x}, \bar{u}, t) \cdot \underline{t_f}$$

"Homotopy" ~~is~~

$$\min J = \Psi(\bar{x}_f, t_f) + \int_{t_0}^{t_f} L(\bar{x}, \bar{u}, t) dt$$

$$\text{subject to: } \dot{\bar{x}} = \bar{f}(\bar{x}, \bar{u}, t)$$

$$\bar{x}(0) = \bar{x}_0 \text{ (given)}$$

$$\bar{\Psi}(\bar{x}_f, t_f) = \bar{0}$$

$$\text{New?} \longrightarrow u_L < \bar{u} < u_u$$

Pontryagin's Maximum Principle (1961)

minimize H over all possible $\bar{u}$.

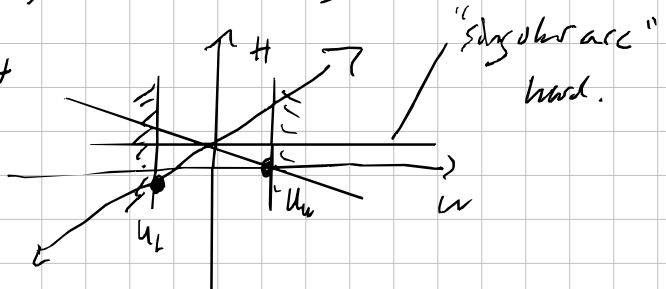
$\rightarrow \min_{\bar{u}} H$ $\bar{u}$ all possible

$$H = L(\bar{x}, \bar{u}, t) + \bar{\lambda}^T \bar{f}(\bar{x}, \bar{u}, t)$$

H linear in terms of $\bar{u}$:
"switching"

$$H = u(\underbrace{C_1}) + (\underbrace{C_2})$$

C are constant



if switchy function

> 0

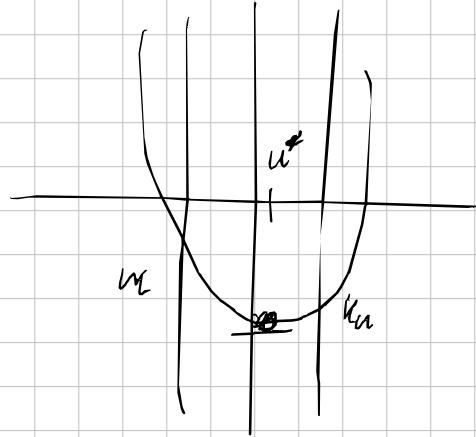
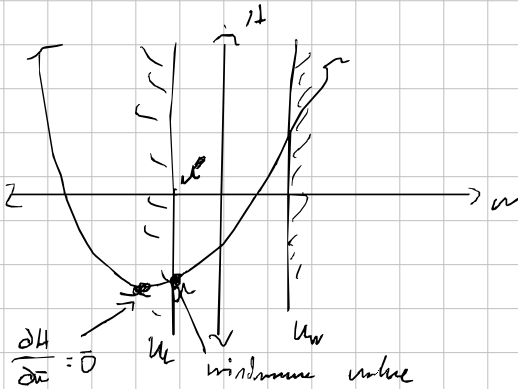
u

u_L

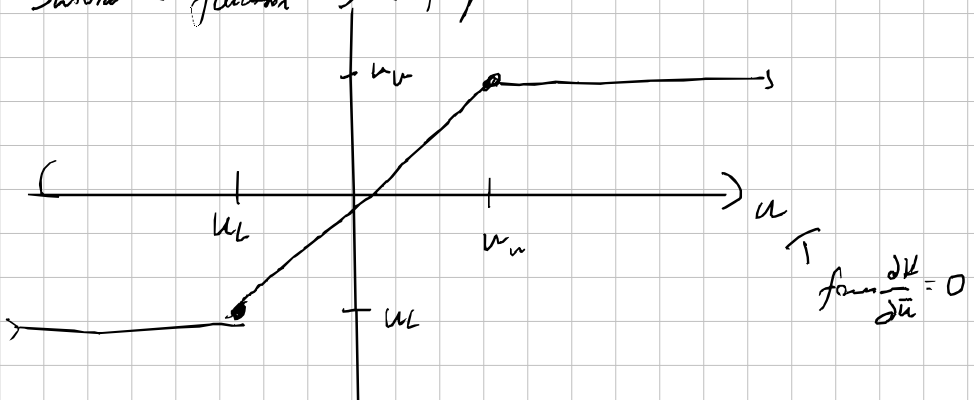
< 0

u_H

OK It is graphed in terms of u : $H = u^2/2 \dots$



"Saturating function" $\rightarrow u^*$, optimal



1D Car Problem

$$\text{min time: } J = \int_0^{t_f} 1 \, dt + \overbrace{\bar{v}^T \bar{\psi}}^{\Phi_f}$$

$$\text{subject to: } \dot{x} = v \quad x(0) = x_0 < 0 \\ \dot{v} = u \quad v(0) = v_0 < 0$$

$$\bar{\psi} = \begin{bmatrix} x(t_f) \\ v(t_f) \end{bmatrix} = \bar{0}$$

$$|u| \leq 1$$

→

$$H = L + \bar{\lambda}^T \bar{f} = 1 + \lambda_x(v) + \lambda_v(u)$$

Linear H in terms of $u \rightarrow u$ must be u_2 or u_1

EL:

$$\dot{\bar{\lambda}}^T = -\frac{\partial H}{\partial \bar{x}} = -[0, \lambda_x] \quad \bar{\lambda}(t_f) = \frac{\partial \Phi_f}{\partial \bar{x}(t_f)} = [v_1, v_2]$$

H linear in terms of u , don't use $\frac{\partial H}{\partial u}$

instead switching function: $s_f = \lambda_v$, anything that works, u is in H .

$$\text{Opt control law: } s_f = \lambda_v < 0, \quad u^* = 1 \\ s_f = \lambda_v > 0, \quad u^* = -1$$

only if $\lambda_v = 0$, $u^* = ?$

Transversality (Lam free) cond.:

$$H(t_f) = - \frac{\partial \Phi}{\partial t_f} = 0$$

$$\dot{\lambda}_x = 0, \lambda_x(t_f) = v_1 \rightarrow \lambda_x(t) = v_1$$

$$\dot{\lambda}_v = -\lambda_x = -v_1, \lambda_v(t_f) = v_2$$

$$\therefore \lambda_v(t) = v_2 + v_1(t_f - t) = S_f \rightarrow \text{linear}$$

↓
max one switch
(zero crossing)

Integrate $\ddot{x}$ from 0 to t only $u^* = +1$

$$\dot{v} = 1 \quad v(0) = v_0$$

$$v(t) = v_0 + t$$

$$\dot{x}(t) = v(t), \quad x(0) = x_0$$

$$x(t) = x_0 + v_0 t + \frac{1}{2} t^2$$

Derive $\dot{x}$ from t_f to t using $u^* = -1$

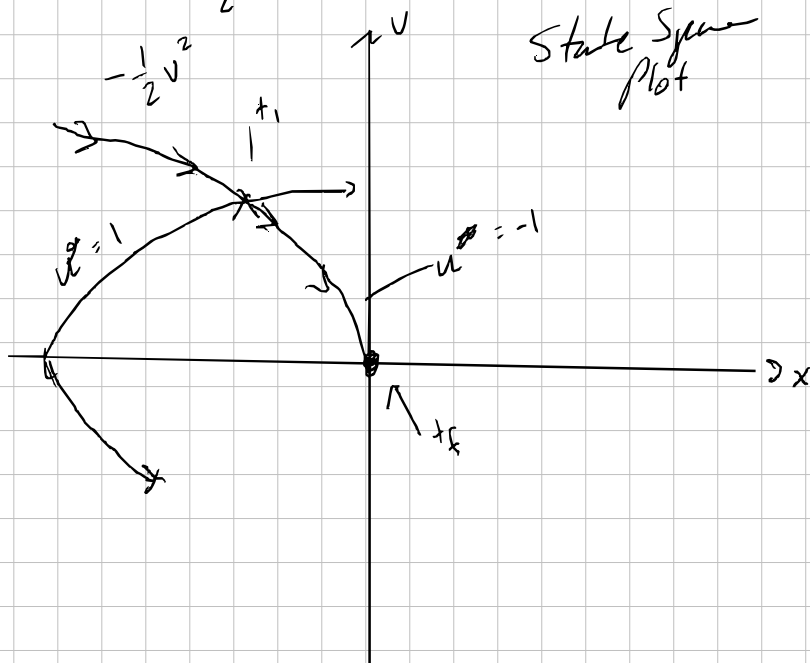
$$\dot{v} = -1, \quad v(t_f) = 0$$

$$v(t) = t_f - t$$

$$\dot{x}(t) = v(t) = t_f - t, \quad x(t_f) = 0$$

$$x(t) = -\frac{1}{2}(t_f - t)^2$$

$$x(v) = -\frac{1}{2}(v)^2$$



$$\Leftrightarrow t_1 : v(t_1) = v_0 + t_1 = t_f - t_1$$

$$x(t_1) = x_0 + v_0 t_1 + \frac{1}{2} t_1^2 = -\frac{1}{2} (t_f - t_1)^2$$

$$t_f = v_0 + 2t_1$$

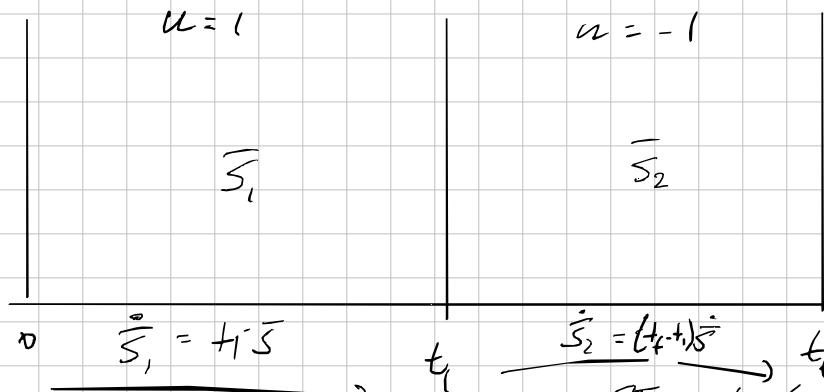
$$x_0 + v_0 t_1 + \frac{1}{2} t_1^2 = -\frac{1}{2} (v_0 + 2t_1 - t_1)^2$$

$$x_0 + v_0 t_1 + \frac{1}{2} t_1^2 = -\frac{1}{2} (v_0^2 + 2v_0 t_1 + t_1^2)$$

$$t_1^2 + 2v_0 t_1 + (x_0 + \frac{1}{2} v_0^2) = 0$$

$$t_1 = \frac{-2v_0 \pm \sqrt{4v_0^2 - 4(x_0 + \frac{1}{2} v_0^2)}}{2}$$

$$t_1 = \frac{-2v_0 \pm \sqrt{2v_0^2 - 4x_0}}{2}$$



$$t = t_1 \tau_1 \leftarrow \tau_1 = \frac{t}{t_1}$$

Acc 1

$$x_1$$

$$v_1$$

$$\lambda x_1$$

$$\lambda v_1$$

$$t_1 \xrightarrow{\bar{s}_2 = (t_f - t) \dot{s}} t_f$$

$$\tau_2 = \frac{t - t_1}{t_f - t_1} \rightarrow t = t_1 + (t_f - t_1) \tau_2$$

Acc 2

$$x_2$$

$$v_2$$

$$\lambda x_2$$

$$\lambda v_2$$

Hamiltonian

$$f_1$$

$$f_f$$

ICS

$$x_f(0) = x_0$$

$$v_f(0) = v_0$$

Continuity Conditions

$$x_1(t_1^-) = x_2(t_1^+)$$

$$v_1(t_1^-) = v_2(t_1^+)$$

$$\lambda x_1(t_1^-) = \lambda x_2(t_1^+)$$

$$\lambda v_1(t_1^-) = \lambda v_2(t_1^+)$$

Terminal BLs

$$x_2(t_f) = 0$$

$$v_2(t_f) = 0$$

Transversality

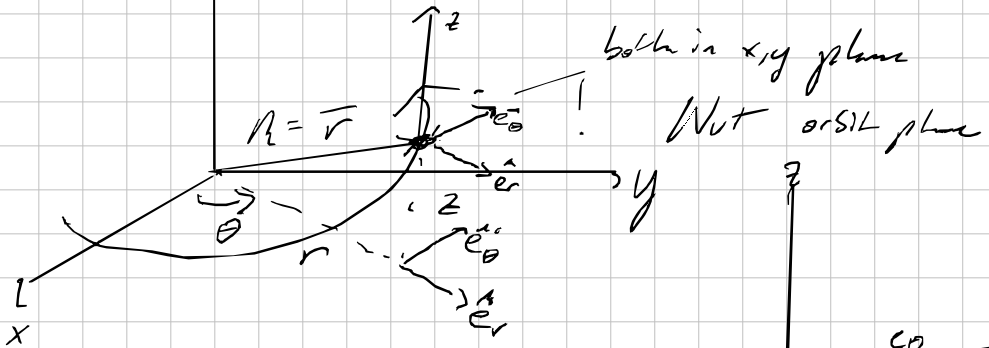
for t_f free, $H(t_f) = 0$

Boundary functions

$$S_f(t_f) = \lambda_v(t_f) = 0$$

$$\dot{S}_1 = \lambda_1 S$$

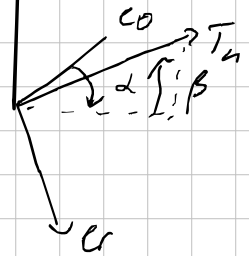
cylindrical coord system



$$R = \sqrt{r^2 + z^2}$$

$$\ddot{\vec{r}} = -\frac{\mu c}{R^3} \vec{r} + \vec{T}_a$$

thrust acceleration



States: r
 θ

$$\sum F_r = m(\ddot{r} - v\dot{\theta}^2) = -\frac{\mu r}{R^3} + S F \cos \beta \sin \alpha$$

switch (1 or 0)

$$\begin{aligned} z \\ v_r = \dot{r} \\ v_\theta = \dot{\theta} \\ v_z = \dot{z} \\ T_a \end{aligned}$$

$$\ddot{r} = r\dot{\theta}^2 - \frac{\mu r}{R^3} + S T_a \cos \beta \sin \alpha$$

$\frac{v_\theta^2}{r}$

Constraints:

$$\frac{dr}{dt} = v_r$$

$$\frac{dz}{dt} = v_z$$

$$\frac{d\theta}{dt} = \frac{v_\theta}{r}$$

$$\frac{dv_r}{dt} = \ddot{r} = \frac{v_\theta^2}{r} - \frac{\mu r}{R^3} + S T_a \cos \beta \sin \alpha$$

$$\sum F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = s \cdot F_{\text{explos}}$$

$$r\ddot{\theta} + \dot{r}\dot{\theta} = -\dot{r}\dot{\theta} + sT_a \cos\beta \cos\alpha$$

$\uparrow$
 $v_r \left(\frac{v_\theta}{r} \right)$

$$\frac{dv_\theta}{dt} = \dot{r}\dot{\theta} + r\ddot{\theta} = -\frac{v_r v_\theta}{r} + sT_a \cos\beta \cos\alpha$$

$$\sum F_z = m\ddot{z} = -\frac{\mu}{R^3} z + s \cdot T_a \sin\beta$$

$$\frac{dv_z}{dt} = -\frac{\mu z}{R^3} + sT_a \sin\beta$$

$$T_a = \frac{F}{m} = \frac{-C_{\text{exh}} \cdot v_{\text{in}}}{m} = \frac{u_{\text{exh}} v_{\text{in}}}{m}$$

$$\frac{dF}{dt} = 0 \quad \therefore F \text{ constant.}$$

$$\frac{dT_a}{dt} = \frac{+C_{\text{exh}} v_{\text{in}}}{m^2} \cdot v_{\text{in}} = s \frac{T_a^2}{C_{\text{exh}}}$$

min fuel: $J = T_a(t_f)$ or $\int_0^t s \, dt$

$$H = s + \lambda_r (v_r) + \lambda_\theta \left(\frac{v_\theta}{r} \right) + \lambda_z (v_z) + \lambda_{v_r} \left(\frac{v_\theta^2}{r} - \frac{\mu}{R^3} + s T_a \cos \beta \sin \alpha \right) \\ + \lambda_{v_\theta} \left(-\frac{v_r v_\theta}{r} + s T_a \cos \beta \cos \alpha \right) + \lambda_{v_z} \left(\frac{-\mu z}{R^3} + s T_a \sin \beta \right) \\ + \lambda_{T_a} \left(s \cdot \frac{T_a^2}{c_{max}} \right)$$

$$\dot{\hat{\lambda}}^T =$$

$$\dot{\hat{\lambda}}^T = - \frac{\partial H}{\partial \bar{x}}$$

$$= - \left[\begin{array}{l} -\lambda_\theta \frac{v_\theta}{r^2} + \lambda_{v_r} \left(-\frac{v_\theta^2}{r^2} - \frac{\mu}{R^3} + \frac{3\mu r^2}{R^5} \right) + \lambda_{v_\theta} \left(\frac{v_r v_\theta}{r^2} \right) + \lambda_z \left(\frac{3\mu z}{R^5} \right) \\ \lambda_{v_r} \left(\frac{3\mu r z}{R^5} \right) + \lambda_{v_z} \left(-\frac{\mu}{R^3} + \frac{3\mu z^2}{R^5} \right) \\ \lambda_r - \lambda_{v_\theta} \left(\frac{v_\theta}{r} \right) \\ \frac{\lambda_\theta}{r} + 2\lambda_{v_r} \left(\frac{v_\theta}{r} \right) - \lambda_{v_\theta} \left(\frac{v_r}{r} \right) \\ \lambda_z \\ \lambda_{v_r} (s \cos \beta \sin \alpha) + \lambda_{v_\theta} (s \cos \beta \cos \alpha) + \lambda_{v_z} (s \sin \beta) + 2\lambda_{T_a} \left(s \cdot \frac{T_a}{c_{max}} \right) \end{array} \right]$$

Opt Control Law.

$$\frac{\partial H}{\partial \bar{u}} = 0 \text{ for } \alpha \text{ and } \beta \text{ (unconstrained)}$$

$$\frac{\partial H}{\partial \alpha} = 0 = \lambda_{v_r} \cdot s \cdot T_a \cdot \cos \beta \cdot \cos \alpha - \lambda_{v_\theta} \cdot s \cdot T_a \cdot \cos \beta \cdot \sin \alpha$$

$$\pm \lambda_{v_\theta} \sin \alpha = \pm \lambda_{v_r} \cos \alpha$$

$$\tan \alpha = \frac{\pm \lambda_{v_r}}{\pm \lambda_{v_\theta}}, \quad H = s T_a \cos \beta \left[\begin{array}{l} \lambda_{v_r} \\ \lambda_{v_\theta} \end{array} \right] \cdot \left[\begin{array}{l} \sin \alpha \\ \cos \alpha \end{array} \right] + \dots$$

$$\rightarrow \tan \alpha = \frac{-\lambda_r}{-\lambda_{v_\theta}}$$

$$H = s T_a \cos \beta \left(- \sqrt{\lambda_{v_r}^2 + \lambda_{v_\theta}^2} \right) \quad \text{minimize antiparallel.}$$

Potential: null for all u .

$$\frac{\partial H}{\partial \beta} = 0 = -\cancel{\lambda_{V_r} \cdot s \cdot T_n \cdot \sin \alpha \sin \beta} - \cancel{\lambda_{V_\theta} \cdot s \cdot T_n \cdot \cos \alpha \cdot \sin \beta} + \lambda_{V_z} \cdot s \cdot T_n \cdot \cos \beta$$

$$\pm \underbrace{(\lambda_{V_r} \sin \alpha + \lambda_{V_\theta} \cos \alpha) \sin \beta}_{-\sqrt{\lambda_{V_r}^2 + \lambda_{V_\theta}^2}} = \pm \lambda_{V_z} \cos \beta$$

$$\tan \beta = \frac{\ominus \lambda_{V_z}}{\oplus \sqrt{\lambda_{V_r}^2 + \lambda_{V_\theta}^2}} ; \quad -\frac{\pi}{2} \leq \beta \leq \frac{\pi}{2}$$

$$\begin{aligned} H &= \lambda_{V_r} \cdot s \cdot T_n \cdot \cos \beta \sin \alpha + \lambda_{V_\theta} \cdot s \cdot T_n \cdot \cos \beta \cos \alpha + \lambda_{V_z} \cdot s \cdot T_n \cdot \sin \beta + \dots \\ &= s \cdot T_n \left[\underbrace{\cos \beta (\lambda_{V_r} \sin \alpha + \lambda_{V_\theta} \cos \alpha)}_{-\sqrt{\lambda_{V_r}^2 + \lambda_{V_\theta}^2}} + \lambda_{V_z} \sin \beta \right] + \dots \end{aligned}$$

$$= s \cdot T_n \begin{bmatrix} -\sqrt{\lambda_{V_r}^2 + \lambda_{V_\theta}^2} \\ \lambda_{V_z} \end{bmatrix} \cdot \begin{bmatrix} \cos \beta \\ \sin \beta \end{bmatrix} + \dots$$

$$= s \cdot T_n \cdot \left(-\sqrt{\lambda_{V_r}^2 + \lambda_{V_\theta}^2 + \lambda_{V_z}^2} \right) + \dots$$

$$0 \leq s \leq 1$$

$$H = s \underbrace{(\lambda_{v_1} \cdot T_h \cdot \cos \beta \cdot \sin \alpha + \lambda_{v_0} T_h \cos \beta \cos \alpha + \lambda_{v_2} T_h \sin \beta + \lambda_h \frac{\dot{\gamma}_h^2}{g_h})}_{S_f}$$

$$S_f = 1 - \sqrt{\lambda_{v_1}^2 + \lambda_{v_0}^2 + \lambda_{v_2}^2} + \lambda_{T_h} \cdot \underbrace{\frac{T_h}{c_{ext}}}_{>0}$$

ExAkr 2:

2 problem

problem setup

what is J ? including terminal cost.

$$\rightarrow J = \varphi(\bar{x}_T, t_T) + \bar{v}^T \bar{\psi} + \int_0^{t_T} L dt$$

EL E_2 undermin-3

$$H = L + \bar{\lambda}^T \bar{f}$$

$$\lambda^T = -\frac{\partial H}{\partial \bar{x}} \quad \lambda^T(t_f) = \frac{\partial \Phi}{\partial x_f}$$

But they are under penalty:
if $\|u\|$ is constrained:

→ It Minimize in terms of u :

i. u is either lower or upper bound
lower or S_f

ii. It is bounded in terms of u ,

→ $\frac{\partial H}{\partial u} = 0$, saturate at lower
bound and upper bound

Or if $\|u\|$ not constrained

$$\frac{\partial H}{\partial u} = 0$$

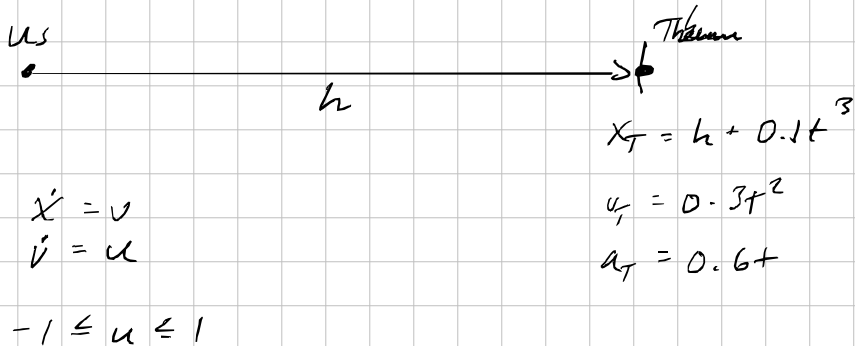
Transversality Condition (from Weier)

$$H(t_f) = -\frac{\partial \Phi}{\partial t_f}$$

if $L + f$ are explicit fns of t ,

$\dot{H} = 0$, H is const for all time.

Exam Problem Problems



minimum time $\rightarrow u = +1$ for all t ($u(t) = 1$)

$\dot{v} = 1$ need to intercept.

$$v(t) = t$$

$$x(t) = \frac{1}{2}t^2$$

$$J = \underbrace{\int_0^{t_f} 1 dt}_{\text{chase}} \quad \text{or} \quad J = t_f$$

$$J = \int_0^{t_f} 1 dt + v[x(t_f) - (h + 0.1t_f^3)]$$

$$L = 1, \quad \Phi = v[x(t_f) - (h + 0.1t_f^3)]$$

EL

$$H = L + \bar{\lambda} \bar{f} = 1 + \lambda_x(v) + \lambda_v(u)$$

$$\left[\dot{\bar{\lambda}}^T = - \frac{\partial H}{\partial \bar{x}} = - [0, \lambda_x] \right]$$

$$\left[\bar{L}(t_f) = \frac{\partial \Phi}{\partial \bar{x}_f} = [v, 0] \right]$$

because u is limited and H is linear in terms of u ,

maximum principle

Pontryagin's Minimum Principle

min H over all possible u

$$H = u(\lambda_v) + \dots$$

$$\frac{\partial H}{\partial u} = 0$$

switching function: $s_f = \lambda_v$

if $\lambda_v < 0$, $u > 0$

$\lambda_v > 0$, $u < 0$

$\lambda_v = 0$, u undefined

all EL eqns.

Transversality cond.:

$$H_f = H(t_f) = - \frac{\partial \Phi}{\partial t_f} = 0.3 v t_f^2$$

$$\lambda_x(t) = v$$

$$\lambda'_v(t) = -\lambda_x = -v \quad \lambda_v(t_f) = 0$$

integrating: $\lambda_v(t) = v(t_f - t)$

λ_v has to be negative.

so $\lambda_v(t) = v(t_f - t) \geq 0$ for all t
 then v is neg for all t .

L and f are not explicit fun of time,
 so $\dot{H}$ is 0, $\therefore H$ is constant.

$$\rightarrow H_f = 0.3v t_f^2 = H(t) \quad (\text{for all } t)$$

transversality
 cond.

1D Car problem

$$\dot{x} = v$$

$$\dot{v} = u$$

fixed final time

min fuel: $J = \int_0^T |u| dt$

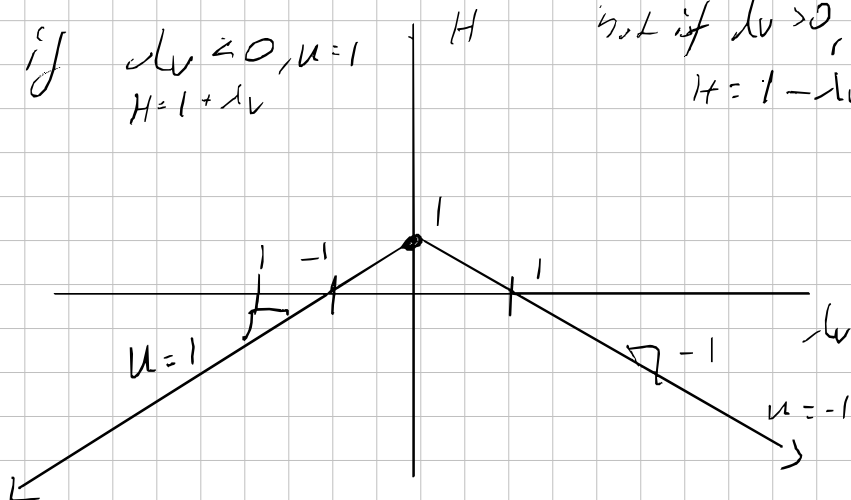
u limited to $-1 \leq u \leq 1$

$$H = |u| + \lambda_x(v) + \lambda_v(u)$$

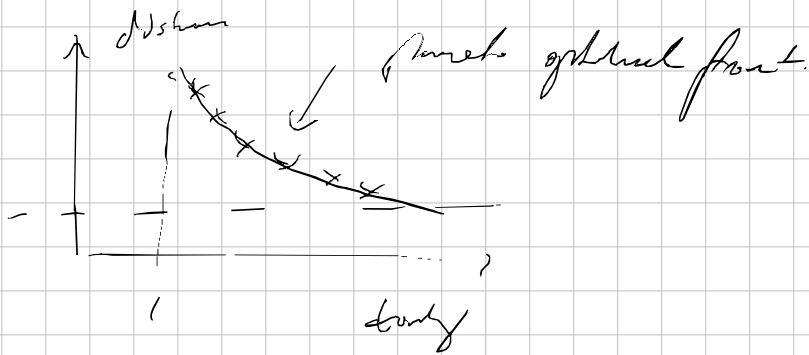
$$H = |u| + \lambda_v(u) + \lambda_x(v)$$

if $\lambda_v \geq 0, u = 1$
 $H = 1 + \lambda_v$

if $\lambda_v < 0, u = -1$
 $H = 1 - \lambda_v$



$$\begin{aligned} \lambda_v < -1, & u = 1 \\ -1 \leq \lambda_v \leq 1, & u = 0 \\ \lambda_v > 1, & u = -1 \end{aligned}$$



In optimality when Lagrangian

Interior Point Constraints

$$\bar{\Psi}_{IP} = 0$$

$$J = \psi_f + \underbrace{\bar{v}_{IP} \bar{\Psi}_{IP}} + \bar{v}_f \bar{\Psi}_f + \int_0^{t_f} L dt$$

$$\bar{\Psi} = \begin{bmatrix} x(t_1) - x_1 \\ y(t_1) - y_1 \end{bmatrix}$$

$$\bar{\lambda}_{t_1^+} = \bar{\lambda}_{t_1^-} + \bar{v}_{IP}^T \frac{\partial \bar{\Psi}_{IP}}{\partial \bar{x}}$$

$$\dot{x} = v \cos \theta$$

$$\dot{y} = v \sin \theta$$

$$\dot{v} = a$$

$$\dot{\theta} = \gamma$$

$$\bar{J} = \frac{1}{r} \int_0^{t_f} a^2 + \gamma^2 dt$$

$$H = \frac{1}{r} (a^2 + \gamma^2) + \lambda_x (v \cos \theta) + \lambda_y (v \sin \theta) + \lambda_1 (a) + \lambda_2 (\gamma)$$

$$\frac{\partial H}{\partial a} = 0 = \frac{2a}{r} + \lambda_1 \Rightarrow a = -\frac{r\lambda_1}{2}$$

$$\dot{\bar{\lambda}} = - \left(\frac{\partial H}{\partial \bar{x}} \right)^T = \begin{bmatrix} 0 \\ 0 \\ -(\lambda_x \cos \theta + \lambda_y \sin \theta) \\ -v (-\lambda_x \sin \theta + \lambda_y \cos \theta) \end{bmatrix}$$

$$\bar{\lambda}^T(t_f) = \frac{\partial \Phi}{\partial \bar{x}_f} = \begin{bmatrix} v_1 \\ v_2 \\ 0 \\ 0 \end{bmatrix}$$

Then is free so we have transversality on

$$H(t_f) = -\frac{d\Phi}{dt} = 0$$

Arc 1

$$\begin{array}{l} \dot{\bar{x}} = \bar{f}(\bar{x}, \bar{u}) \\ \dot{\bar{\lambda}} = -\frac{\partial H}{\partial \bar{x}} \\ t_1 \end{array} \left. \vphantom{\begin{array}{l} \dot{\bar{x}} = \bar{f}(\bar{x}, \bar{u}) \\ \dot{\bar{\lambda}} = -\frac{\partial H}{\partial \bar{x}} \\ t_1 \end{array}} \right\} \begin{array}{l} \text{8 eqns} \\ \text{9 variables total} \end{array}$$

Arc 2

$$\begin{array}{l} \dot{\bar{x}} = \bar{f}(\bar{x}, \bar{u}) \\ \dot{\bar{\lambda}} = -\frac{\partial H}{\partial \bar{x}} \\ t_2 \end{array} \left. \vphantom{\begin{array}{l} \dot{\bar{x}} = \bar{f}(\bar{x}, \bar{u}) \\ \dot{\bar{\lambda}} = -\frac{\partial H}{\partial \bar{x}} \\ t_2 \end{array}} \right\} \begin{array}{l} \text{8 eqns} \\ \text{9 variables total} \end{array}$$

Need 18 constraints.

Initial Con

$$x(0) = 10$$

$$y(0) = 10$$

$$v(0) = 0$$

$$\lambda_0(0) = -\frac{\partial E_0}{\partial x} = 0$$

Continuity Eqs

$$\bar{x}(t_i^-) = \bar{x}(t_i^+) \quad \text{3 eqns}$$

$$\begin{array}{l} \text{Ls } x(t_i^-) = x_i \\ y(t_i^-) = y_i \\ \lambda_v(t_i^-) = \lambda_v(t_i^+) \\ \lambda_0(t_i^-) = \lambda_0(t_i^+) \end{array} \left. \vphantom{\begin{array}{l} x(t_i^-) = x_i \\ y(t_i^-) = y_i \\ \lambda_v(t_i^-) = \lambda_v(t_i^+) \\ \lambda_0(t_i^-) = \lambda_0(t_i^+) \end{array}} \right\} \begin{array}{l} \text{city coord.} \\ \text{4 eqns} \end{array}$$

Terminal cons

$$x(t_2) = x_2$$

$$y(t_2) = y_2$$

$$d_v(t_2) = 0$$

$$d_\theta(t_2) = 0$$

Transversality cons

$$H(t_2) = 0$$

$\rightarrow \mathcal{L} + \bar{F}$ not explicit fns of time $\therefore \dot{H} = 0 \rightarrow H(t) = 0$

$$H(t_1) = 0$$

$$\tau_k = \frac{t}{t_k}$$

$$d\tau_k = \frac{1}{t_k} dt$$

$$\frac{d\bar{x}}{d\tau} \cdot \frac{d\tau}{dt} = \frac{d\bar{x}}{dt} = F(\bar{x}, a) \cdot t_k$$

Hamilton's Principle

minimize $J = \int_0^{+f} \mathcal{L} E + U_{1 \rightarrow 2} dt$

From dynamics: $\underbrace{U_{1 \rightarrow 2}}_{\text{conservative}} = -\Delta PE = PE_1 - PE_2$

$$J = \int_0^{+f} \underbrace{\mathcal{L} E - PE}_{\text{Lagrangian}} + \underbrace{(U_{1 \rightarrow 2})_{\text{non}}}_{\text{non conservative work}} dt$$

$\int \vec{F}_{\text{non}} \cdot d\vec{r} + \int \vec{m}_{\text{cm}} \cdot d\vec{\theta} = \int \vec{Q}_{\text{non}} \cdot d\vec{q}$

$$J = \int_0^{+f} \mathcal{L} + (U_{1 \rightarrow 2})_{\text{non}} dt$$

$$J = \int \left(\mathcal{L} + \int_{\vec{q}_0}^{\vec{q}_f} \vec{Q}_{\text{non}} \cdot d\vec{q} \right) dt$$

Subject to: some equations of motion

$$\ddot{\vec{q}} = \vec{f}(\vec{q}, \vec{u}, t) = \vec{u}$$

some control variables $\vec{u}$ are
don't know f

$$H = \mathcal{L} + \int_{\vec{q}_0}^{\vec{q}_f} \vec{Q}_{\text{non}} \cdot d\vec{q} + \vec{\lambda}^T \vec{u}$$

$$\mathcal{L} = \mathcal{L}(\vec{q}, \dot{\vec{q}}, \vec{u}, t)$$

$$\dot{\lambda}^T = -\frac{\partial H}{\partial \bar{q}} = -\left(\frac{\partial \mathcal{L}}{\partial \bar{q}} + \bar{Q}_{non}\right)$$

$$\frac{\partial H}{\partial \bar{u}} = 0 = \frac{\partial \mathcal{L}}{\partial \bar{u}} + \bar{\lambda}^T$$

$$\bar{\lambda}^T = -\frac{\partial \mathcal{L}}{\partial \bar{u}} = -\frac{\partial \mathcal{L}}{\partial \dot{\bar{q}}}$$

$\underbrace{\hspace{10em}}_{\frac{d}{dt}}$

$$\dot{\lambda}^T = -\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\bar{q}}} \right) = -\frac{\partial \mathcal{L}}{\partial \bar{q}} - \bar{Q}_{non}$$

$$\boxed{\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\bar{q}}} \right) - \frac{\partial \mathcal{L}}{\partial \bar{q}} = \bar{Q}_{non}} \quad \text{Lagrangian EOM}$$

Generalized non conservative forces.

What is the Hamiltonian actually?

$$H = \mathcal{L} + \int \bar{Q}_{non} \cdot d\bar{q} + \left(-\frac{\partial \mathcal{L}}{\partial \dot{\bar{q}}} \right) \left(\dot{\bar{q}} \right)$$

$$= KE - PE \quad \rightarrow \quad KE = \frac{1}{2} m \dot{\bar{q}} \cdot \dot{\bar{q}} = \frac{1}{2} \dot{\bar{q}}^T [I] \dot{\bar{q}}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\bar{q}}} = m \dot{\bar{q}} + [I] \dot{\bar{q}}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\bar{q}}} \dot{\bar{q}} = m \dot{\bar{q}}^T \dot{\bar{q}} + \dot{\bar{q}}^T [I] \dot{\bar{q}} = 2KE$$

$$H = KE - PE + \int \bar{Q}_{non} \cdot d\bar{q} - 2KE$$

$$H = -KE - PE + \int \bar{Q}_{non} \cdot d\bar{q}$$

iff conservative system $\rightarrow 0$

$$H = -KE - PE = -\text{Energy} = \text{constant.}$$

$$\text{Ex: } \bar{q} = \begin{Bmatrix} x \\ \theta \end{Bmatrix} \quad \dot{\bar{q}} = \begin{Bmatrix} v \\ \omega \end{Bmatrix}$$

Dynamic Programming

Let the following be posed

$$\min J(\bar{x}_0, t_0) = \varphi(\bar{x}_f, t_f) + \int_{t_0}^{t_f} L(\bar{x}, \bar{u}, t) dt$$

$$\text{subject to: } \dot{\bar{x}} = \bar{f}(\bar{x}, \bar{u}, t)$$

$$\text{w/ terminal cost: } \bar{\psi}(\bar{x}_f, t_f) = 0$$

$$J(\bar{x}, t) = \varphi(\bar{x}_f, t_f) + \int_t^{t_f} L(\bar{x}, \bar{u}, \tau) d\tau$$

@ future time, $t + \Delta t$ and future states $\bar{x} + \Delta \bar{x}$

$$J(\bar{x}, t) = \underbrace{\varphi(\bar{x}_f, t_f)}_{\text{terminal cost}} + \underbrace{\int_t^{t+\Delta t} L(\bar{x}, \bar{u}, \tau) d\tau + \int_{t+\Delta t}^{t_f} L(\bar{x} + \Delta \bar{x}, \bar{u}, \tau) d\tau}_{\text{running cost}}$$

$$J(\bar{x} + \Delta \bar{x}, t + \Delta t)$$

$$J(\bar{x}, t) = J(\bar{x} + \Delta \bar{x}, t + \Delta t) + \int_t^{t+\Delta t} L(\bar{x}, \bar{u}, \tau) d\tau$$

Optimal cost from current step to end

$$J^*(\bar{x}, t) = \min_{\substack{\bar{u}(\tau) \\ t \leq \tau \leq t+\Delta t}} \left[\int_t^{t+\Delta t} L(\bar{x}, \bar{u}, \tau) d\tau + J^*(\bar{x} + \Delta \bar{x}, t + \Delta t) \right]$$

Taylor series expansion:

$$J^*(\bar{x} + \Delta\bar{x}, t + \Delta t) \approx J^*(\bar{x}, t) + \underbrace{\frac{\partial J^*}{\partial \bar{x}} \Delta\bar{x} + \frac{\partial J^*}{\partial t} \Delta t}_{\text{H.O.T}}$$

$$J^*(\bar{x}, t) = \min_{\substack{\bar{u}(\tau) \\ t \leq \tau \leq t + \Delta t}} \left[\int_t^{t+\Delta t} L(\bar{x}, \bar{u}, \tau) d\tau + J^*(\bar{x} + \Delta\bar{x}, t + \Delta t) \right]$$

~~$$J^*(\bar{x}, t) = \min_{\bar{u}(\tau)} \left[\int_t^{t+\Delta t} L(\bar{x}, \bar{u}, \tau) d\tau + J^*(\bar{x}, t) + \frac{\partial J^*}{\partial \bar{x}} \Delta\bar{x} + \frac{\partial J^*}{\partial t} \Delta t \right]$$~~

not fun of $\bar{u}$

$$-\frac{\partial J^*}{\partial t} \Delta t = \min_{t \leq \tau \leq t + \Delta t} \bar{u}(\tau) \left[\int_t^{t+\Delta t} L(\bar{x}, \bar{u}, \tau) d\tau + \frac{\partial J^*}{\partial \bar{x}} \Delta\bar{x} \right]$$

first order approx: $L(\bar{x}, \bar{u}, t) \Delta t$

$$\begin{aligned} \frac{\Delta\bar{x}}{\Delta t} &= \dot{\bar{x}} \\ &= f(\bar{x}, \bar{u}, t) \\ \Delta\bar{x} &= f(\bar{x}, \bar{u}, t) \Delta t \end{aligned}$$

$$-\frac{\partial J^*}{\partial t} = \min_{t \leq \tau \leq t + \Delta t} \bar{u}(\tau) \left[L(\bar{x}, \bar{u}, t) + \frac{\partial J^*}{\partial \bar{x}} f(\bar{x}, \bar{u}, t) \right]$$

Hamilton - Jacobi - Bellman
 (HJB) equations

$$H(\bar{x}, \bar{u}, \lambda, t)$$

$$-\frac{\partial J^*}{\partial t} = \min_{t \leq \tau \leq t + \Delta t} \bar{u}(\tau) \left[H(\bar{x}, \bar{u}, \frac{\partial J^*}{\partial \bar{x}}, t) \right]$$